Benzene

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Benzene

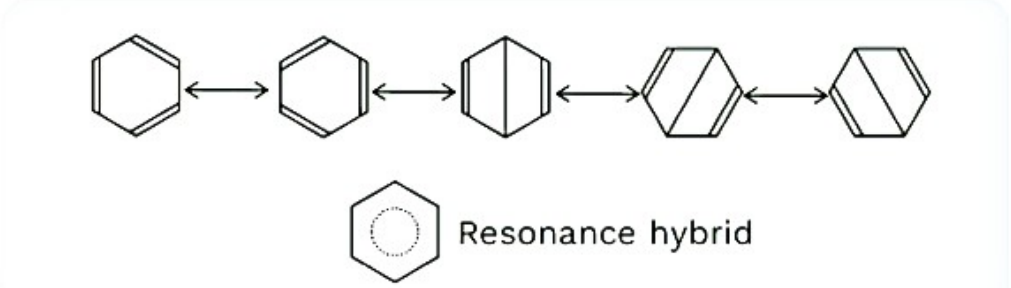
Main Source

Fractional Distillation of Coal Tar : Different fractions are :

Temperature	Main constituents
80° – 170°C	Benzene , toluene, xylene etc.
170° – 230°C	Naphthalene, Phenol etc.
230° – 270°C	Cresols, naphthalene
270° – 360°C	Anthracene
Residue	Carbon

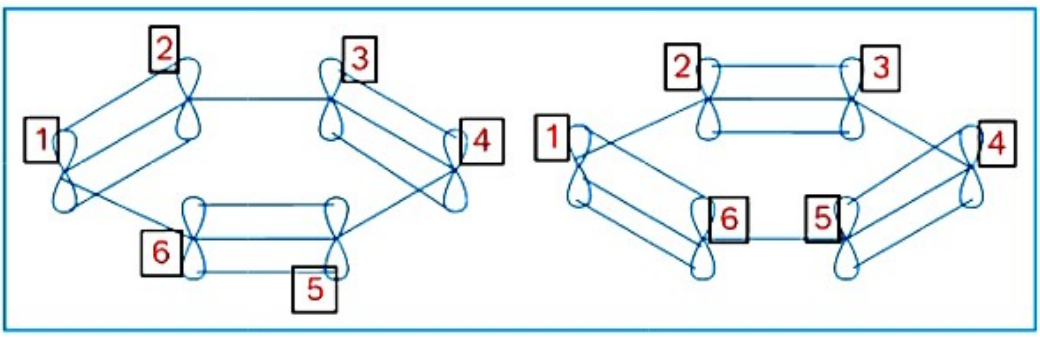
Benzene

(i) Discoverer :Michael Faraday
(ii) C–hybridisation : sp^2
(iii) Geometry–Hexagonal
(iv) Bond angle : 120°
(v) C–C Bond length : 1.39Å
(vi) C–H Bond length : 1.09Å
(vii) Resonance in benzene :



(viii) Number of angles of 120° : 18
(x) Localized and delocalized π–orbital

Localized π–orbital Localized π–orbital



Partial overlapping of p–orbital C_1–C_2, C_3–C_4 and C_5–C_6	Partial overlapping of p–orbital C_2–C_3, C_4–C_5 and C_1–C_6

Obviously, each orbital formed in the partial overlapping is attracted by two carbon nucleus hence less stable.

Let's Understand

Based on its structure and properties, what kind of reaction should benzene undergo?

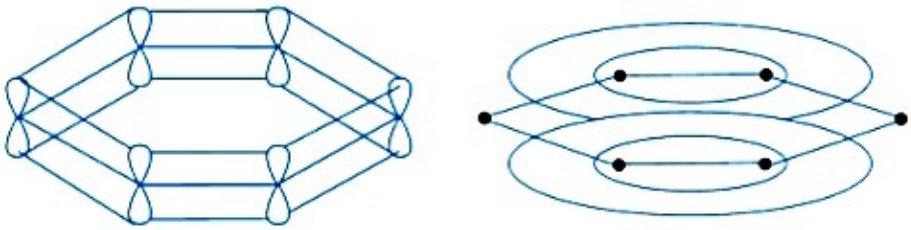
Are any of its bond particularly weak?

Does it have electron rich or electron deficient atom?

Electrophilic Aromatic Substitution (EAS)

Electron rich benzene ring, having delocalized electron in six R-orbitals that overlap above and below the plane of ring.
Stable benzene ring, obey huckeli's rule, (4n+2) $\pi e^{(-)}$ rule stable the ring, Only favor specific kind of reactions.

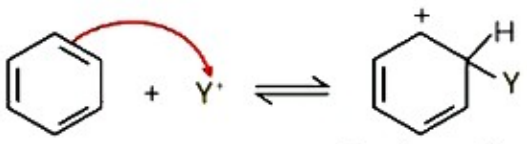
Delocalized P-orbital:



Electron cloud above and below the plane of benzene shields the ring carbon from the attack of a nucleophile.

Reaction Mechanism of benzene

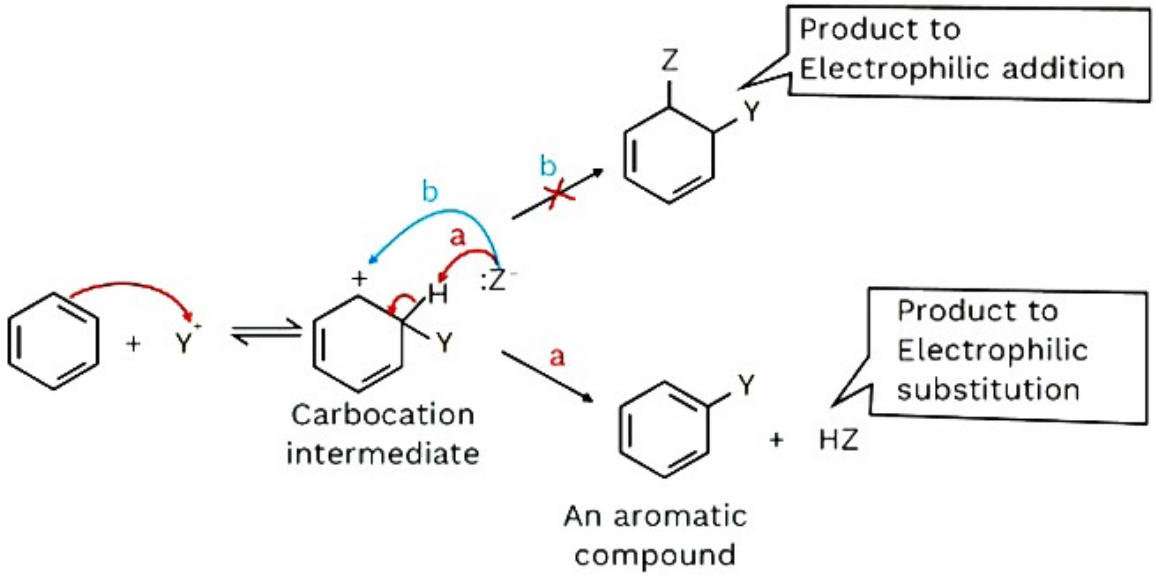
Nucleophiliclty of benzene (π electron cloud above and below plane) favour reaction with electrohile. Nucleophilic benzene when attaches with electrophile (E^+), carbocation is intermediate.



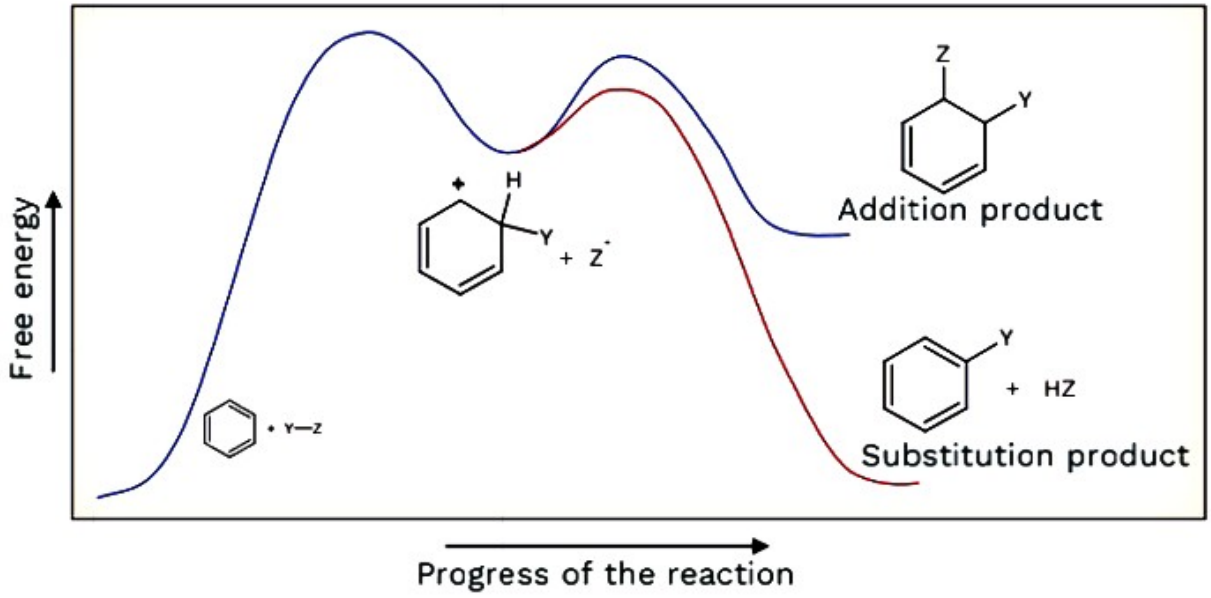
Carbocation

intermediate

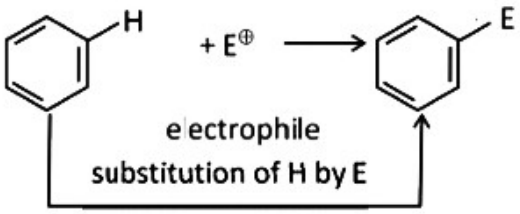
- This should remind you of the first step in an electrophilic addition reaction of an alkene



- If the carbocation intermediate formed from the reaction of benzene with an electrophile were to react similarly with a nucleophile, the addition product would not be aromatic.
- If the carbocation loses a proton from the site of electrophilic attack, the aromaticity of the benzene ring is restored. Because the aromatic product is much more stable than the non-aromatic addition product, the overall reaction is an electrophilic substitution reaction rather than an electrophilic addition reaction.
- In the substitution reaction, an electrophile substitutes for one of the hydrogens attached to the benzene ring.
- Benzene undergoes electrophilic substitution reactions that preserve aromaticity, rather than electrophilic addition reaction (the reactions characteristic of alkenes), which would destroy aromaticity.



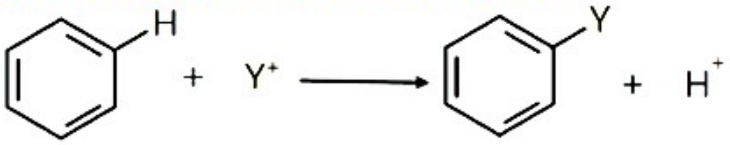
As a result, the characteristic reaction of benzene is electrophilic aromatic substitution- A hydrogen atom is replaced by an electrophile.



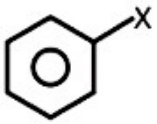
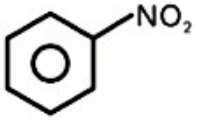
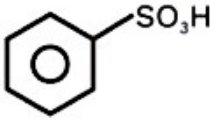
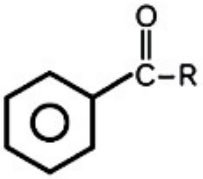
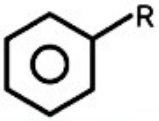
Aromatic Substitution Reactions:

Benzene shows electrophilic substitution reaction in which electrophile substitute hydrogen on Aromatic Ring so that's why it is called as electrophilic Aromatic Substitution Reaction.

Electrophilic Aromatic Substitution Reaction (EAS)



In case of benzene, the most important electrophilic substitution reactions are described as

Reaction	$E^{\oplus}$ (Electrophile)	Product
Halogenation	X^+ (Br, Cl, I)	
Nitration	NO_2^+	
Sulfonation	$\overset{\oplus}{S}O_3H$	
Friedel-Craft Acylation	$R-\overset{\oplus}{C}=O$	
Friedel-Craft Alkylation	$\overset{\oplus}{R}$	

The General Mechanism of EAS

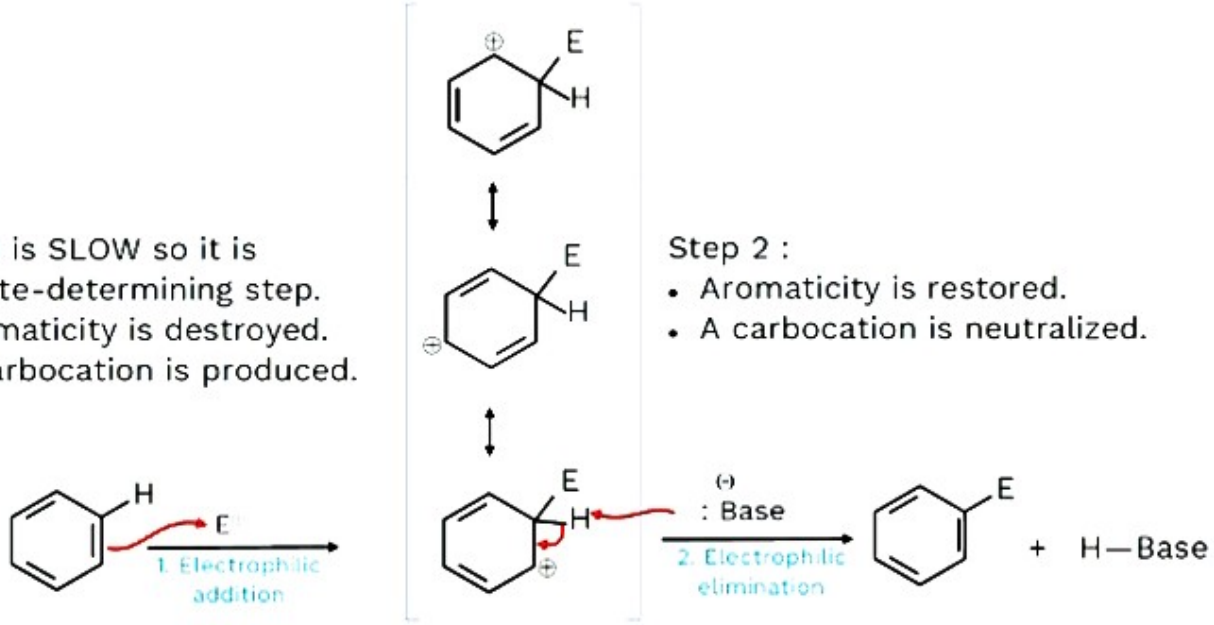
EAS reactions occur via a two steps mechanism.

Step 1. Addition of the electrophile

General Mechanism for electrophilic aromatic substitution:

Step 1 is SLOW so it is the rate-determining step.

- Aromaticity is destroyed.
- A carbocation is produced.



Step 2 :

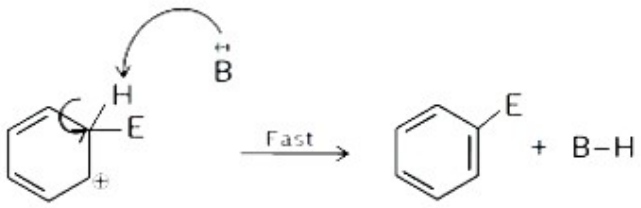
- Aromaticity is restored.
- A carbocation is neutralized.

Arenium ion intermediate

Resonance stabilized

In setp-1 the intermediate is formed and it is called as arenium ion/Wheland ion, in which five sp2 and one sp3 hybridise carbon atom are present.

Step 2. This intermediate have three resonating structure and all have same energy so any of them will be used to form product.



Addition of Electrophile on benzene ring is slow step i.e. first step due to aromaticity will break so this first step is consider as transition state and rate determining step. Energy will be high in compare to second steps.
In second step removal of H^+ (proton) with the help of conjugate base, ring will attain Aromaticity and get substituted by electrophile (E^+).

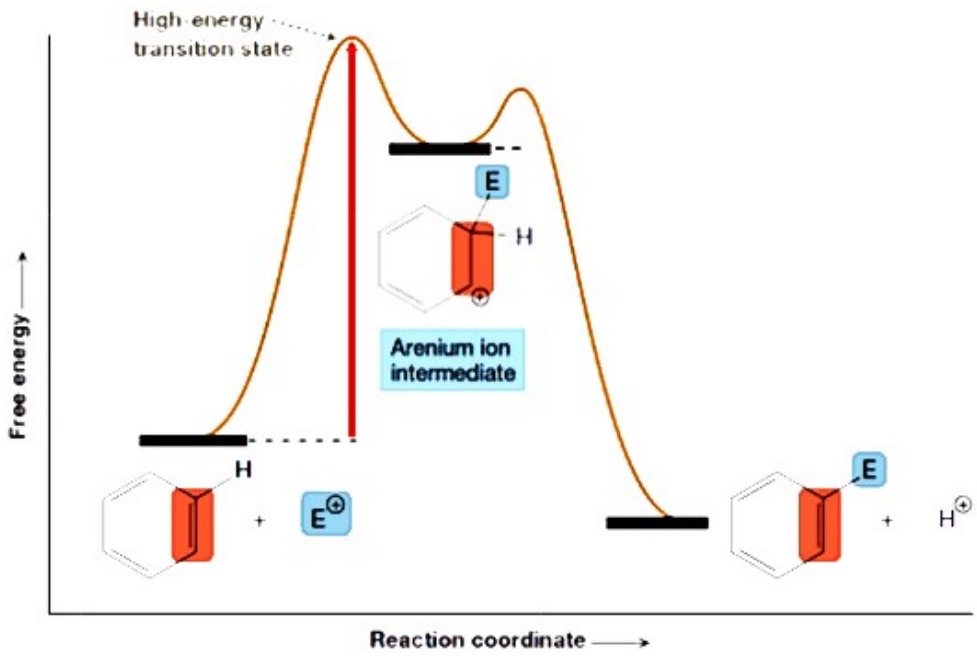
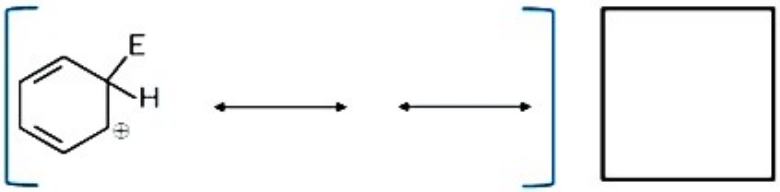


Figure : Reaction energy diagram for an electrophilic aromatic substitution.

Points to remember

The arenium ion intermediate is significantly less stable than either the reactants or products, due to the loss of aromaticity and the formation of a positively charged C that lacks an octet. As a result, Step 1 has a very-high-energy transition state and is the slow step.

Draw all of the resonance structures and the resonance hybrid of the arenium ion intermediate. Include the curved arrows that show how each resonance structure is converted to the next one. In the resonance hybrid, how many C atoms share the positive charge?



Hybrid

Because benzene and the electrophile both appear as reactants in the rate determining step, the rate of the overall reaction depends on the concentration of both species, as indicated

$$\text{Rate} = K \left[\text{C}_6\text{H}_6 \right] \left[E^{+} \right]$$

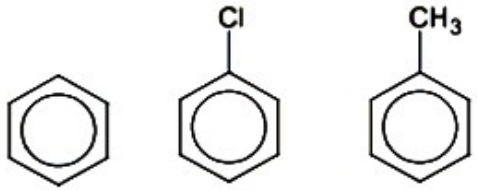
Rate of electrophilic substitution at aromatic benzene or derivatives depends upon stability of sigma complex :

$$\text{Rate} \propto +M \text{ or } +H \text{ or } +I$$

$$\propto -\frac{1}{M} \text{ or } -\frac{1}{H} \text{ or } -\frac{1}{I}$$

Q. From the given compounds find out rate of electrophilic aromatic substitution

Q232



(i) (ii) (iii)

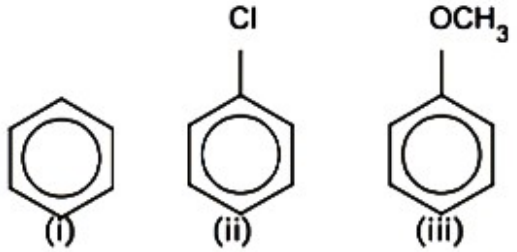
Sol.

(iii) > (i) > (ii)

→ Electron donating groups Rate.

- CH_3 : + HC (Increase rate w.r.t Benzene)
- Cl : –I (decrease rate w.r.t benzene)

Q233

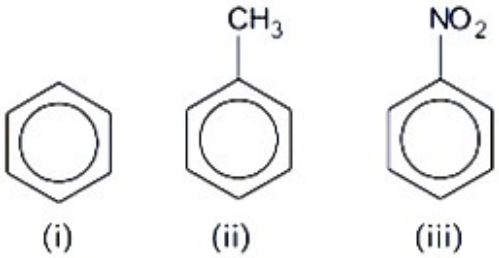


Sol. (iii) > (i) > (ii)

$-CH_3$: +M group

$-Cl$: −I group

Q234

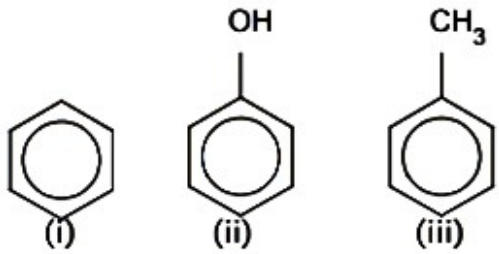


Sol. (ii) > (i) > (iii)

$-CH_3$: +HC group

$-NO_2$: −M group

Q235

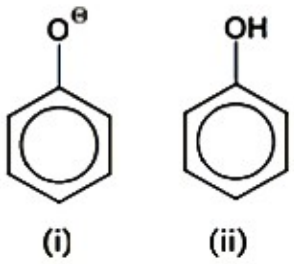


Sol. (ii) > (iii) > (i)

$-OH$: +M group -(+M groups increase rate more than + H and +I groups)

$-CH_3$: +HC group

Q236

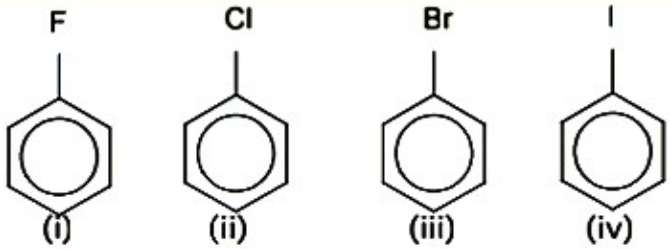


Sol.

(i) > (ii)

+M effect of $-O^{\ominus}$ > $-OH$

Q237

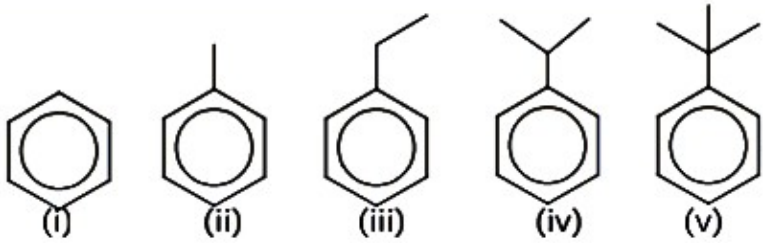


Sol.

(iv) > (iii) > (ii) > (i)

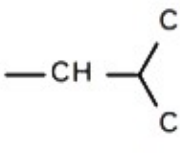
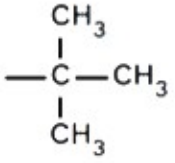
As halogens are – I group and –I effect order: –F > –Cl > –Br > – I

Q238



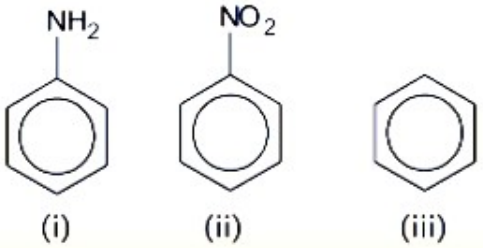
Sol.

(ii) > (iii) > (iv) > (v) > (i)

+ H effect of $-CH_3 > -C_2H_5 >$  > 

Number of ∝-H

Q239



(i) (ii) (iii)

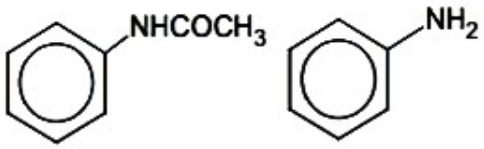
Sol.

(i) > (iii) > (ii)

– NH_2 : +M effect ⇒ Increases rate w.r.t benzenes

– NO_2 : –M effect ⇒ Decreases rate w.r.t benzenes

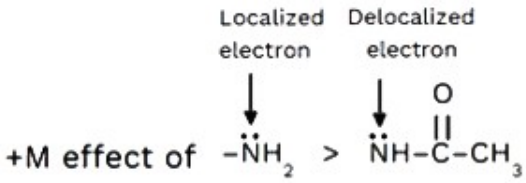
Q240



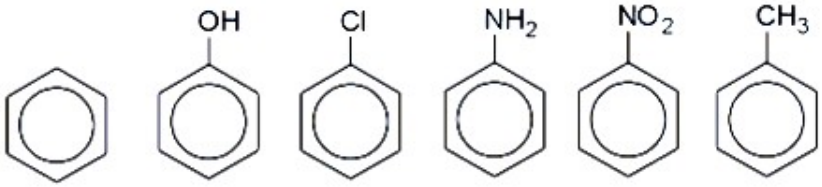
(i) (ii)

Sol.

(ii) > (i)



Q241



(i) (ii) (iii) (iv) (v) (VI)

Sol.

(iv) > (ii) > (vi > (i) > (iii) > (v)

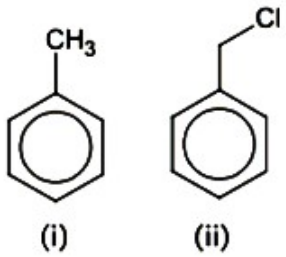
+M : $-NH_2$ > $-OH$

+H : $-CH_3$

–I : –Cl

–M : $-NO_2$

Q242

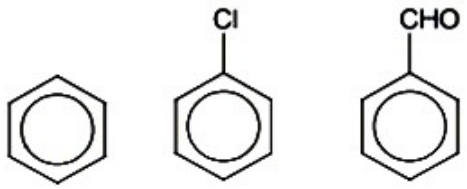


(i) (ii)

Sol.

(i) > (ii)
–CH_3 : + H group
–CH_2Cl : – I group

Q243

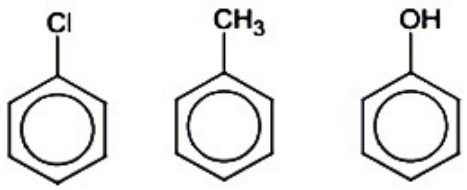


(i) (ii) (iii)

Sol.

(i) > (ii) > (iii)
–Cl : –I group
–CHO : –M group

Q244

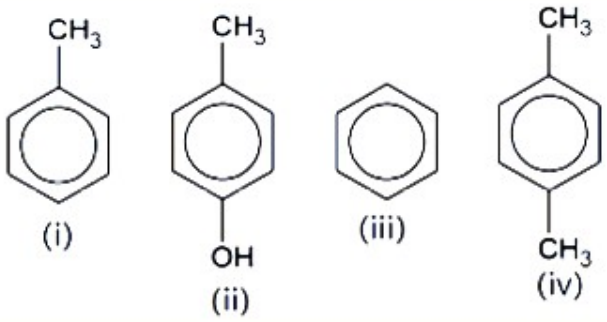


(i) (ii) (iii)

Sol.

(iii) > (ii) > (i)
–OH : + M group
$-CH_3$: + H group
–Cl : – I group

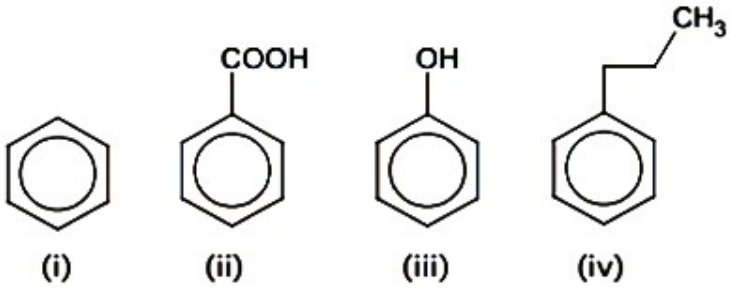
Q245



Sol.

(ii) > (iv) > (i) > (iii)
(i) → [$-CH_3$: + H]
(ii) → [$-CH_3$: +H], [–OH : +M]
(iv) → [Two CH_3^- + H groups]

Q246



Sol.

(iii) > (iv) > (i) > (ii)
(ii) → [–COOH : –M group]
(iii) → [–OH : +M group]
(iv) → [$-CH_2CH_2CH_3$: + H → + I group]

Tips and Trick

The order of electrophilic aromatic substitution is first order with respect to benzene and electrophile Concentration.

$$(Rate)_{EAS} = K[C_6H_6][E^+]$$

Q247 **Suppose that the concentration of the electrophile is doubled and the concentration of benzene is tripled in an electrophilic aromatic substitution reaction. What happens to the rate of the overall reaction?**

Sol. Rate increases 6 times

As rate = K[Benzene] [Electrophile]

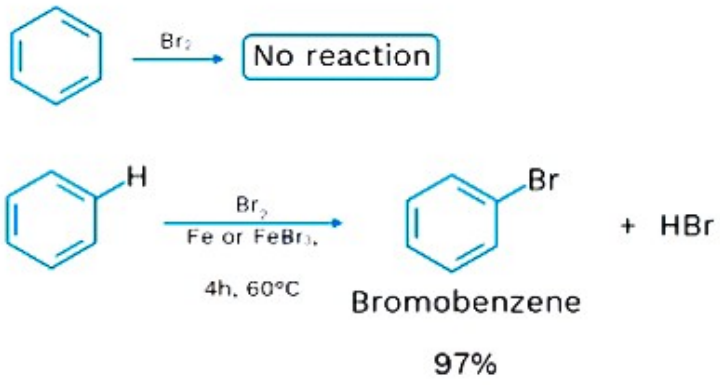
$R_1 = K[C_6H_6]$ [Electrophile]

$R_2 = K\ 3[C_6H_6] \times 2$ [Electrophile]

$= K \times 6\ [C_6H_6] \times$ [Electrophile]

$R_2 = 6\ R_1$

Halogenation



$2\ Fe + 3\ Br_2 \longrightarrow 2\ FeBr_3$

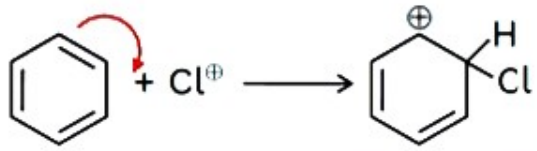
$2\ Fe + 3\ Cl_2 \longrightarrow 2\ FeCl_3$

Introduction

Recall from the Introduction that benzene does not react with molecular chlorine, Cl_2, under normal conditions. In the presence of Fe or $FeCl_3$, however, **chlorination** occurs to yield chlorobenzene; this is a type of **aromatic halogenation** reaction. Similarly, molecular bromine, Br_2, does not react with benzene under normal conditions, but **bromination** produces bromobenzene in the presence of Fe or $FeBr_3$.

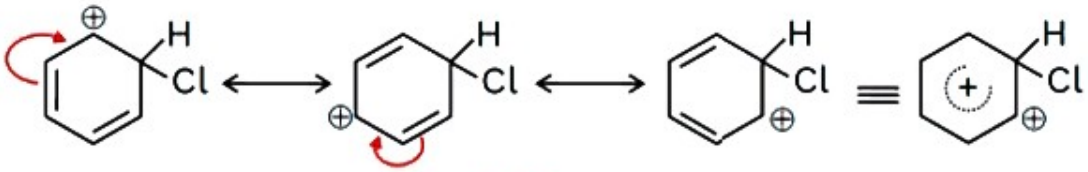
Low temperature and the presence of a halogen carrier favour nuclear substitution. The chlorides or bromides of Al, Fe, Sb may be used:

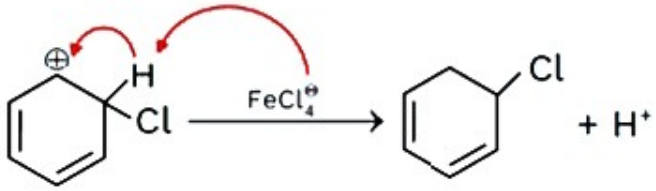
$$Fe \xrightarrow{Cl_2} FeCl_2 \overset{Cl_2}{\rightleftharpoons} FeCl_4^- + Cl^+$$



sigma-complex

(stable by resonance)





$$FeCl_4^- + H^+ \longrightarrow FeCl_3 + HCl$$

Points to remember

Iron is most commonly used being converted to Lewis's acid $FeCl_3$ as shown above. It is again S_E reaction and without halogen-carrier (Lewis's acid), Cl^+ (halonium) is not formed and hence, reaction is not possible:

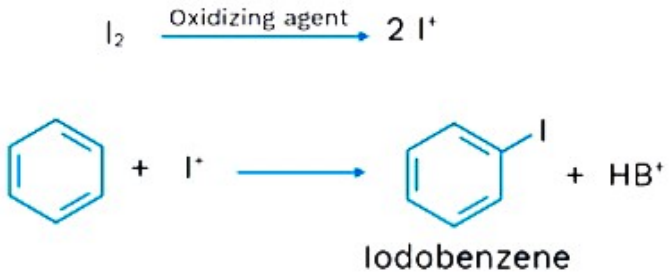
In Electrophilic Aromatic Substitution reaction, bromination and chlorination require lewis Acid to convert Br_2 and Cl_2 in electrophilie, while in case of alkene Lewis Acid not require due to high reactivity then benzene results Br_2 and Cl_2 easily attack on Alkene without convert Electrophile.

Bromination and chlorination also occur by $\overset{\oplus}{H}O\overset{\oplus}{Br}$ and $\overset{\oplus}{H}O\overset{\oplus}{Cl}$, obtained by reaction between H_2O and X_2.

$H_2O + X_2 \rightarrow 2HOX$ (X = Cl, Br)

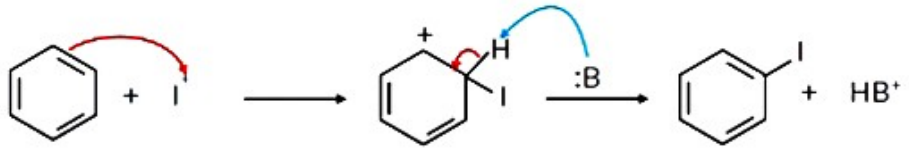
In iodination, I^+ is formed by reaction of HNO_3 with I_2, Nitric oxide act as oxidizing agent here.

Iodination:

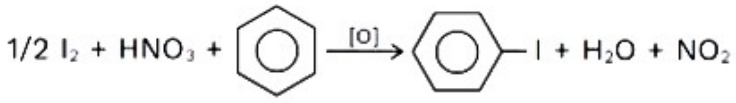


As formation of electrophile in iodination reaction, it will follow same mechanism of Halogenation (bromination or chlorination).

Mechanism for Iodination :



Direct iodination is not possible since, iodine is least reactive and HI formed makes reaction reversible. In presence of oxidising agents like HNO_3 or HIO_3, iodination of benzene is possible and HI formed is converted to I_2.

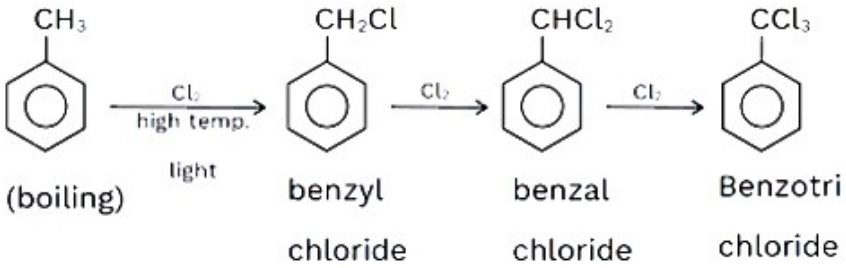


$$5\ HI + HIO_3 \longrightarrow 3\ I_2 + 3H_2O$$

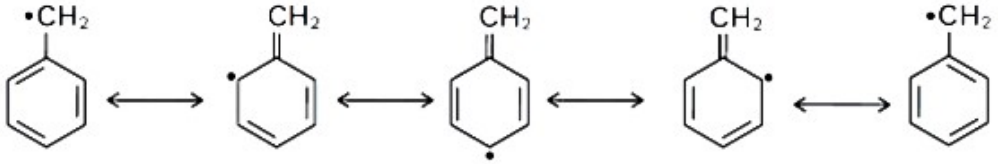
Points to remember

Iodination probably involves an electrophilic aromatic substitution with iodonium (I^+) acting as the electrophile. I^+ is formed from iodine by oxidation with HNO_3.

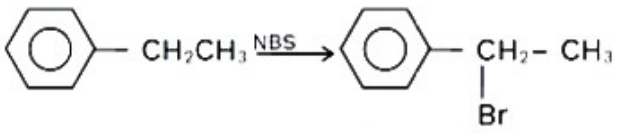
Direct Halogenation
Side-Chain Derivatives



Side-chain halogenation involves free radical mechanisms due to lower bond energy of the benzyl C–H bond :
Benzyl radical is stabilised by resonance due to delocalisation of odd electrons into ring :

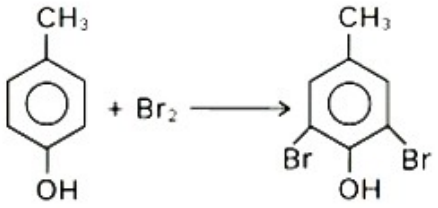


NBS can also be used for bromination of side-chain alkyl group.



(by allylic substitution)

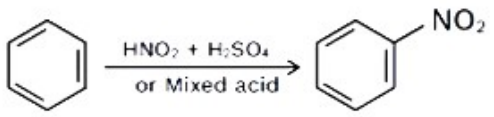
In presence of an activating group–OH or $-NH_2$, side chain halogenation is difficult:



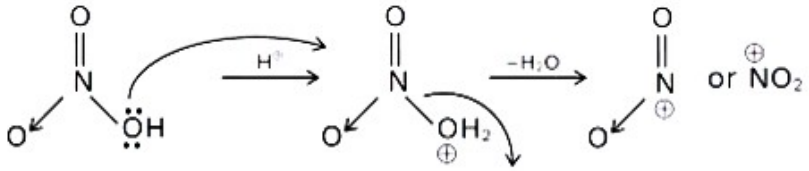
Nitration of Benzene :

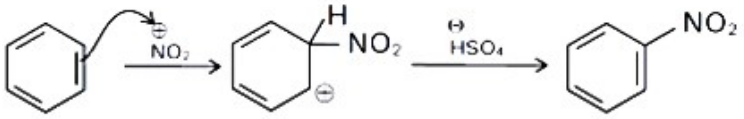
The introduction of a nitro ($-NO_2$) group on benzene known as nitration of benzene. The most common reagent for the nitration of aromatic compound is mixture of concentration HNO_3 and concentration H_2SO_4 (mixed acid).

The electrophile required for the nitration is nitronium ion ($-\overset{\oplus}{N}O_2$).



$$H_2SO_4 \rightleftharpoons H^{\oplus} + \bar{H}SO_4$$





Points to remember

- Besides mixed acid, other reagents have also been used for nitration such as nitronium salts

$$\overset{\oplus}{N}O_2\,\overset{\ominus}{Cl},\ \overset{\oplus}{N}O_2\overset{\ominus}{F},\overset{\oplus}{N}O_2,\overset{\ominus}{Cl}O_4,\overset{\oplus}{N}O_2,\overset{\oplus}{S}O_3$$

- In the case of fuming HNO_3 alone $\overset{\oplus}{N}O_2$ is formed as follow''''

$$2HNO_3 \rightleftharpoons \overset{\oplus}{N}O_2 + \overset{\oplus}{N}O_3 + H_2O$$

- In non-polar solvent N_2O_5 get dissociated and produces $\overset{\oplus}{N}O_2$,

$$N_2O_5 \rightleftharpoons \overset{\oplus}{N}O_2 + \overset{\oplus}{N}O_3$$

Q248 **Compare rate of nitration among C_6H_6, C_6D_6 and C_6T_6.**

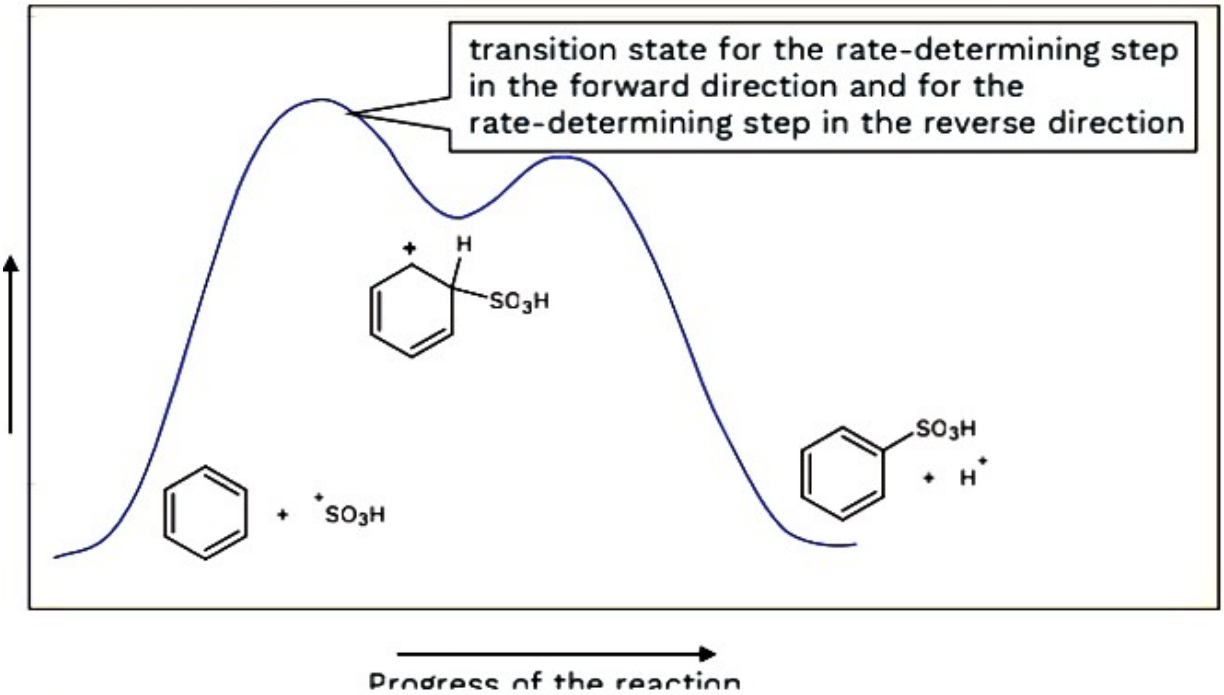
Sol.

$C_6H_6 = C_6D_6 = C_6T_6$

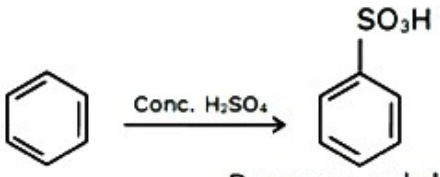
Rate depends on sigma complex formation

Therefore nitration of benzene does not shows primary kinetic isotopic effect.

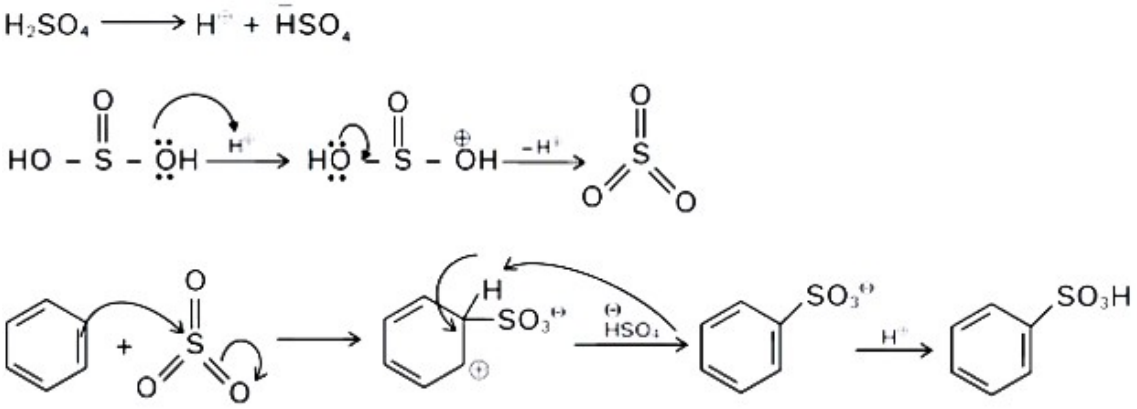
Sulphonation of Benzene :



Sulphonation is usually done with concentrationH_2SO_4.Electrophile required for sulphonation reaction is SO_3.

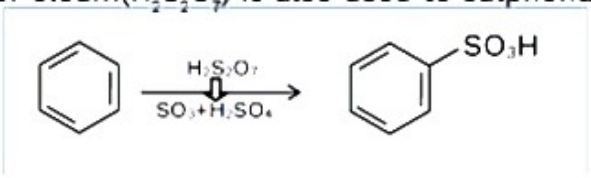


Benzene sulphonic acid

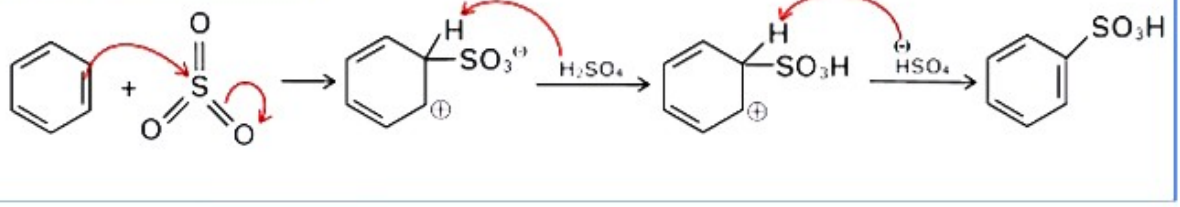


Second method

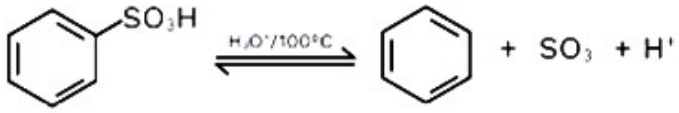
Fuming sulphuric acid or oleum($H_2S_2O_7$) is also used to sulphonate aromatic ring.



Mechanism



As Benzene Sulphonic Acid heated with dilute acid like H_2SO_4 then reaction move backward so that sulphonation reaction is reversible.



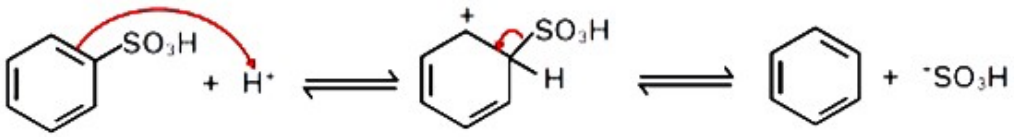
The **principle of microscopic reversibility**

According to microscopic observation every step of mechanism which is reversible only possible when the intermediate is same and reaction trace exact path as defined in forward reaction . Reversiblity concept in reaction of sulponation is one kind of its example . Iodination is also example of reversibility concept.

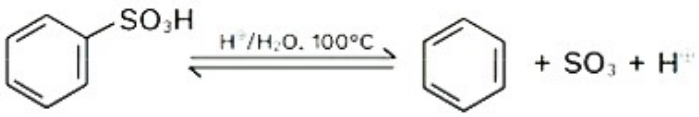
Example:

Reaction coordinate diagram of sulphonation, the forward reaction is sulphonation and backward reaction is di sulphonation along with same intermediate.

Mechanism for desulphonation :



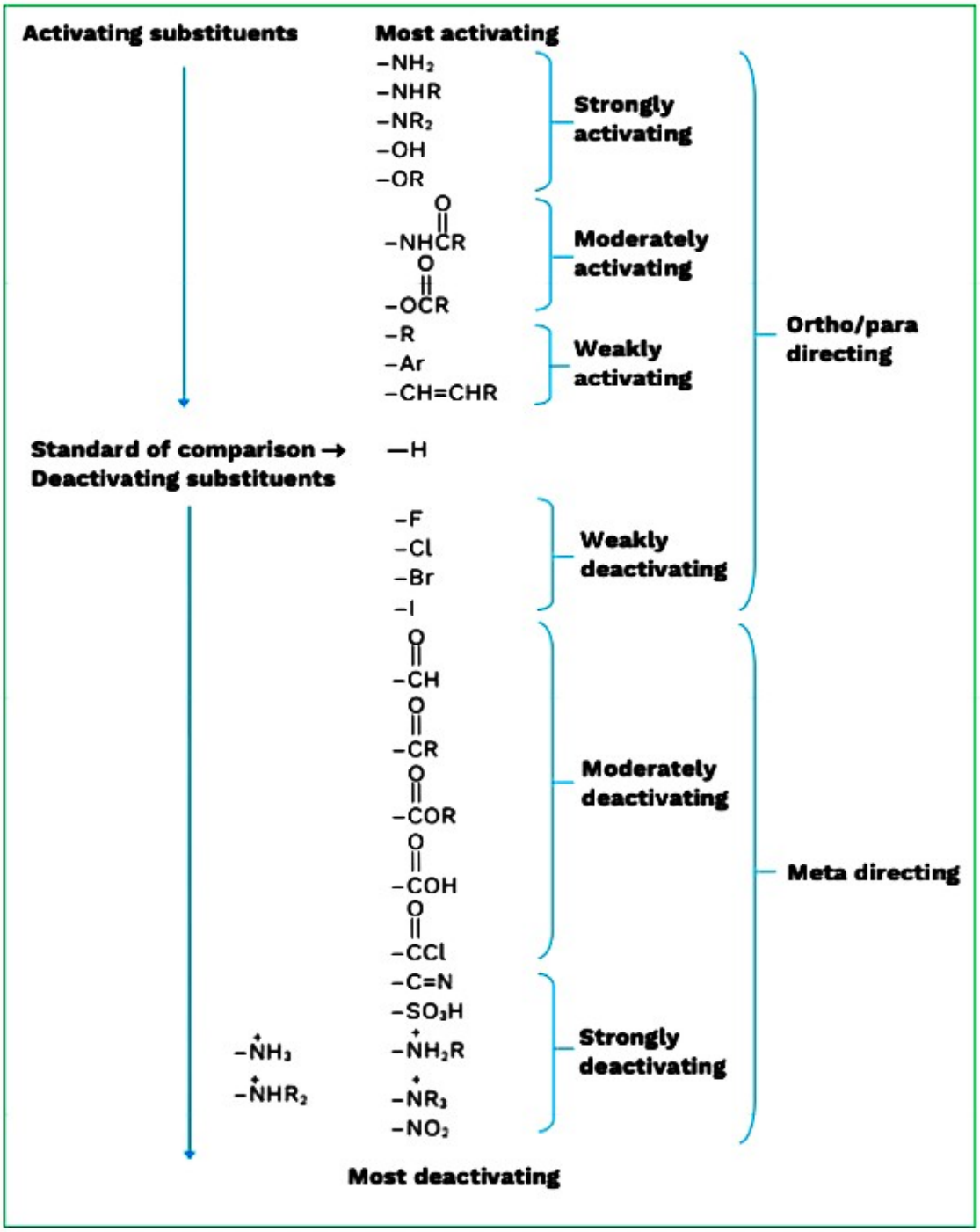
Sulphonation of benzene is a reversible reaction. If benzenesulphonic acid is heated in dilute acid, the reaction proceed in reverse direction.



Due to reversibility in reaction shows kinetic isotopic effect.
Rate of reaction → C–H > C–D > C–T

Relative Reactivity of Substituted Benzene:

The effects of substituents on the reactivity of a Benzene ring and Electrophilic substitution

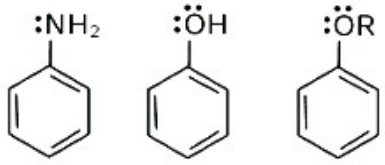


Mechanisms of the above reactions have been given in further section.

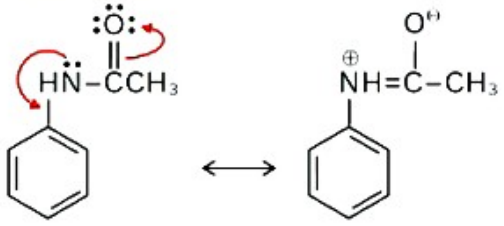
1. All meta-directing groups are deactivating groups;
2. All ortho, para-directing groups except for the halogen are activating groups.
3. The halogens are deactivating groups

   Activated group make benzene more reactivity towards electrophile because activating group incase e—density of benzene ring while deactivating group decrease the reactivity of benzene ring.

   Strong activators apply +m, +I and donate electrons to benzene ring and increases electron density of ring so electrophile attack easily.

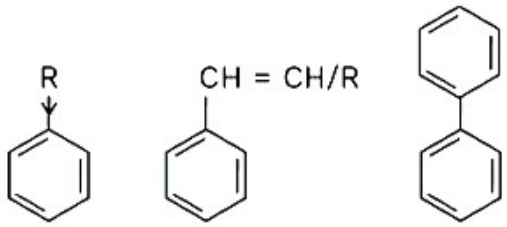


Moderately activators also donate electron but due to self resonance like $-NH-CH=CH_2$, $-O-CH=CH_2$, $-NH-COR$ they also having capacity to withdraw electron from ring.

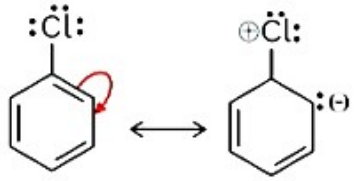


(substituent's donates electrons always from the benzene ring)

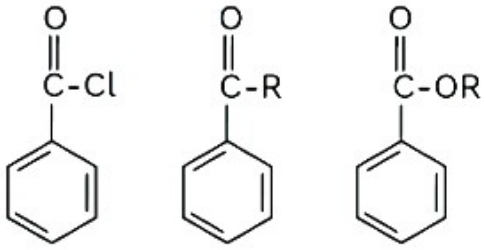
Weakly activating groups like Alkyl (–R), Alkene ($-CH=CH_2$) donate electron as well as withdraw electrons due to resonance but according practical observation these groups are slightly electron donating in net result so act as weakly activators.



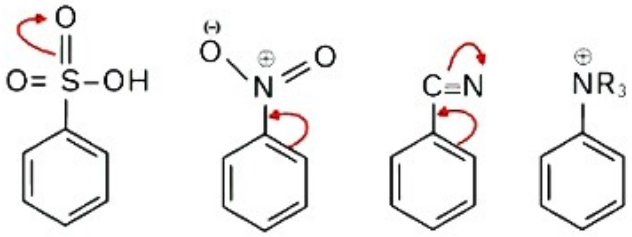
Halogen are weak deactivators due to exceptionally inductive effect is powerful as compared to resonance. This is due to high electronegativity of Halogens.



Moderately deactivators are mainly –CHO, –COR in which carbon is directly connected with ring so they have withdrawing tendency due to resonance & inductive effect.

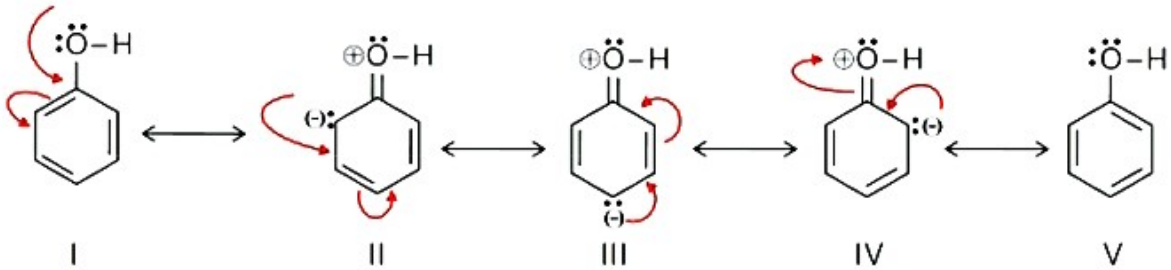


Strong deactivator like $-NO_2$, $-\overset{\oplus}{N}R_3$, $-\overset{\oplus}{N}H_3$ are most powerful withdrawing capacity due to –M or –I effect. Even $NH_4^{\oplus}$ ammonium ion also act as withdrawing tendency due to positive charge.

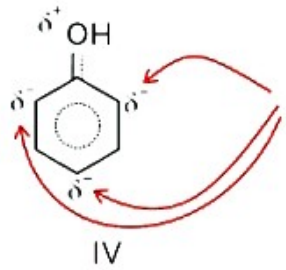


Directing influence as decided by electron density :

–OH group is ortho and para-directing since it increase electron-density at ortho and para-positions for electrophilic attacks; electron-density at meta-position is decreased hence, electrophilic attack at this position is least.



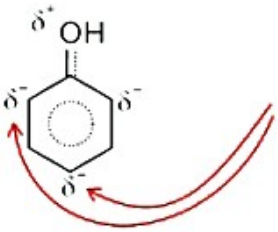
1 to V are different resonance structures and VI resonance hybrid structure.



IV

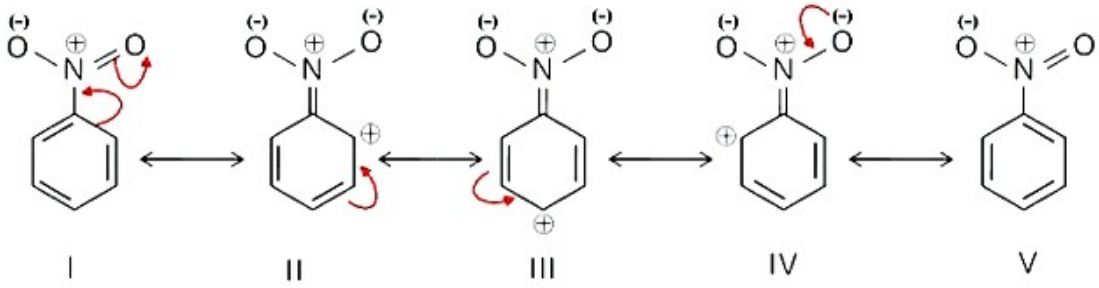
Electrophile can attack the sites with increased electron density which are ortho para w.r.t. –OH group

Naturally nucleophile cannot attack ortho and para-sites but can attack atmeta-site (but less reactive)



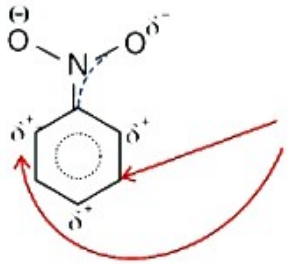
Nucleophile can attack meta-sites

$-NO_2$ group is electron-withdrawing group. It deactivates ortho and para-position for electrophilic attack and thus electrophile can occupy meta-position. I to V are resonance and VI is resonance hybrid.



I II III IV V

Electrophile can attack the sites with increased electron density which are ortho para w.r.t. –OH group

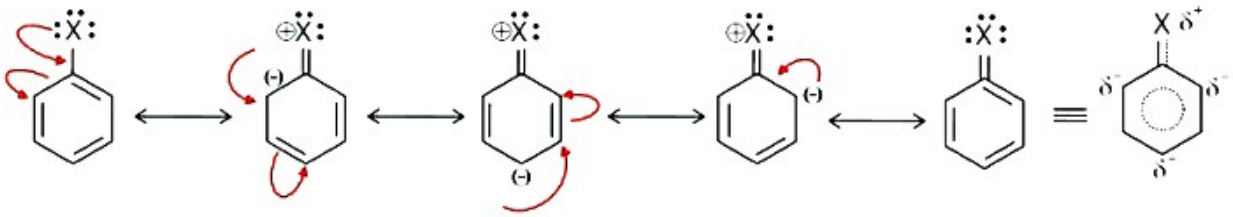


In this case nucleophilic attack at electron-rich ortho and para-sites is favoured.

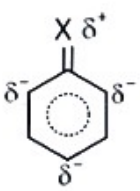


Anomalous Behaviour of halogen substituent :

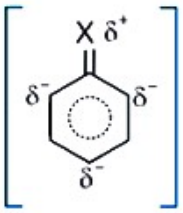
– X(F, Cl, Br, I) is o–, p-directing but is **deactivating** group.



This deactivating nature is attributed to high electronegativity of the halogen atom due to which they withdraw electrons; resonance effect explains its electron-donating behaviour.



(+M↓, – I↑)

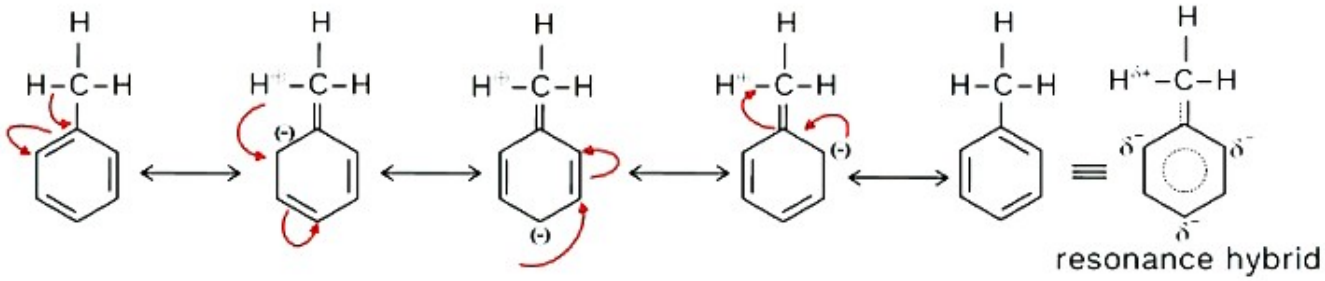


electron displacement due to inductive effect not compensated by resonance effect

overall result; o–, p-directing but deactivating

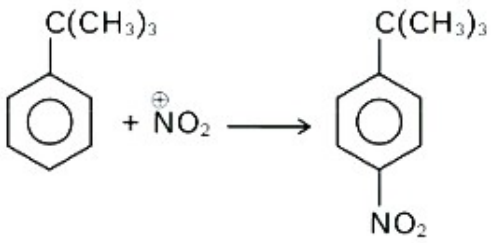
Alkyl groups and Orientation

Alkyl groups ($-CH_3$, $-CH_2CH_3$, etc.) have complete octet and are without one-pair of electrons; but they are ring activators and o-, p-directors. They not only show electron-pumping effect, but also when attached to conjugated system show hyperconjugated effect (no-bond resonance). This type of resonance requires the interaction of σ-electrons instead of usual lone-pair of electrons. In toluene, hyperconjugation is shown below:



Thus, hyperconjugation helps in increasing-density at o-and p-sites in case of alkyl groups.

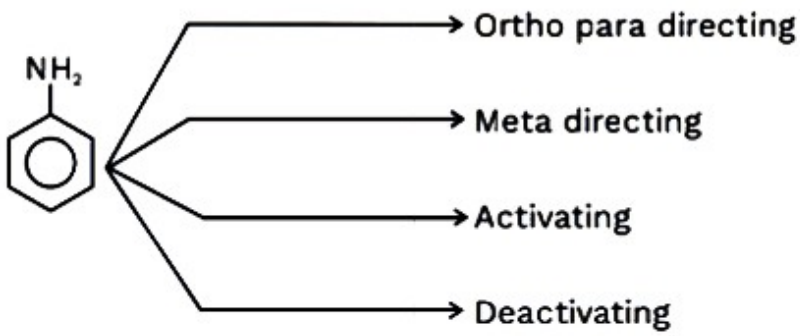
There must be at least one H-atom on the carbon of the alkyl group joined to the ring carbon to take part in hyperconjugation and relative ability of alkyl groups for this effect is : methyl > 1° > 2° > 3°



In this case, only p-isomer is obtained. It is due to steric influence of the substituent's $-C(CH_3)_3$ and due to its unability to show hyperconjugation.

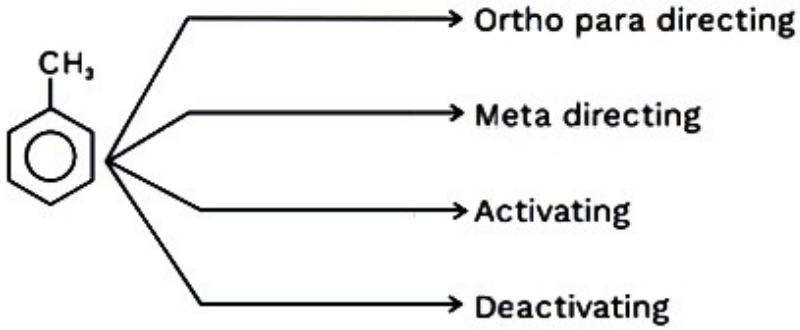
Q. From the given compounds find out rate of electrophilic aromatic substitution

Q249



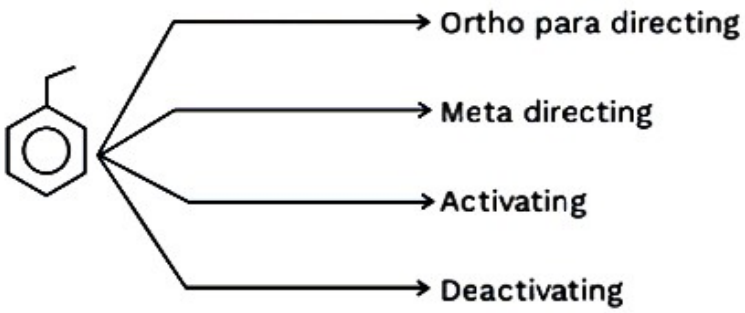
Sol. Ortho para directing (+M group), Activating

Q250



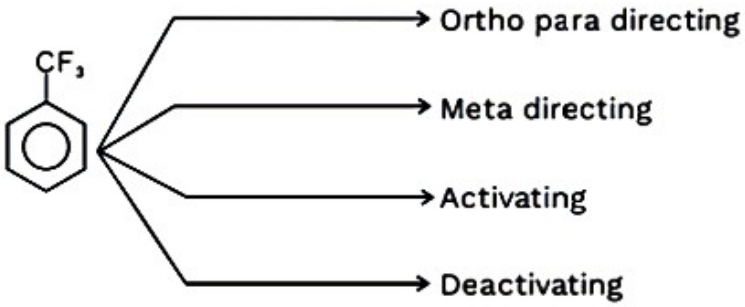
Sol. Ortho para directing (+H group), Activating

Q251



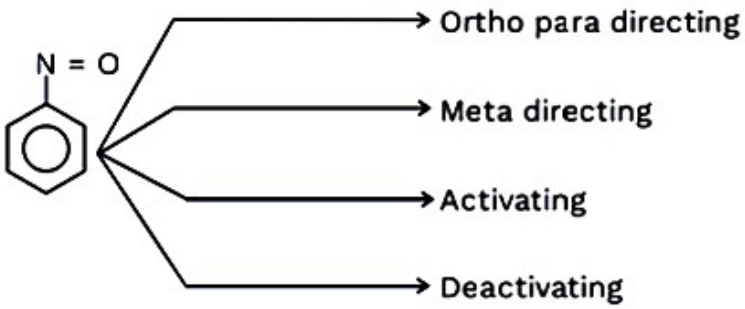
Sol. Ortho para directing (+H group), Activating

Q252



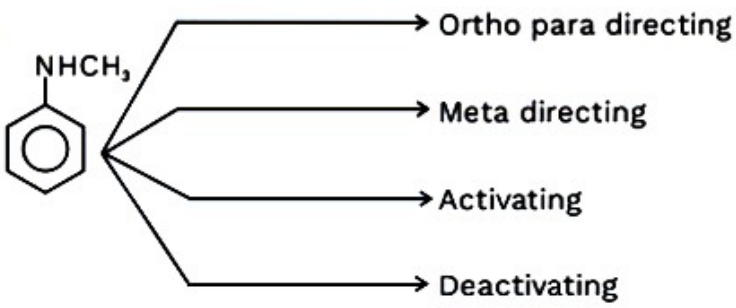
Sol. Meta directing (–I group), Deactivating

Q253



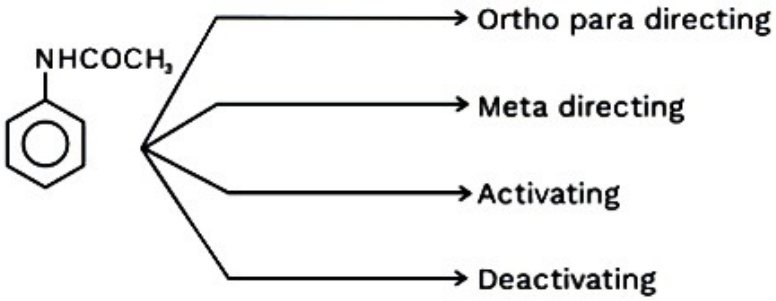
Sol. Ortho para directing (–m group), Deactivating

Q254



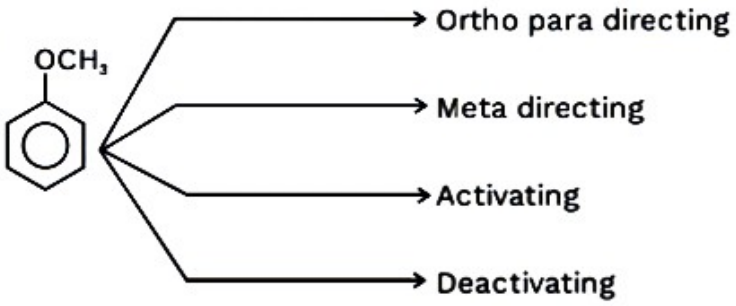
Sol. Ortho para directing (+M group), Activating

Q255



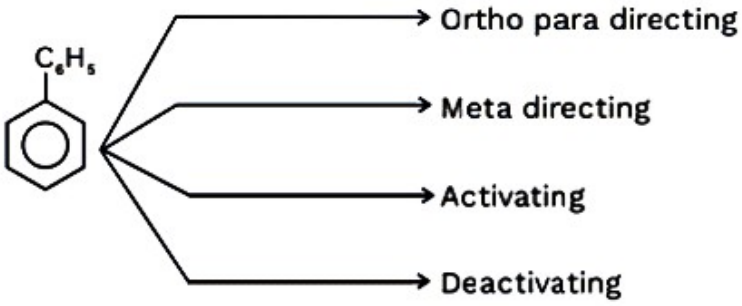
Sol. Ortho para directing (+M group), Activating

Q256



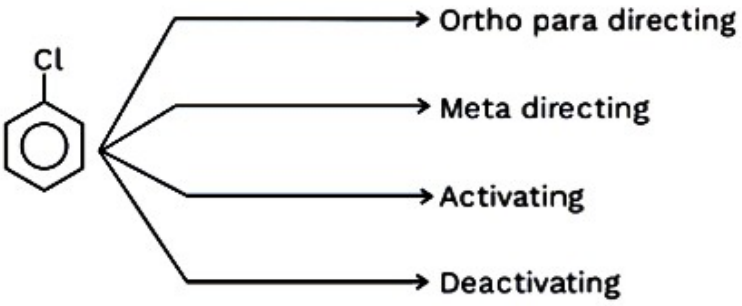
Sol. Ortho para directing (+M group), Activating

Q257



Sol. Meta director directing (–I group), Deactivating

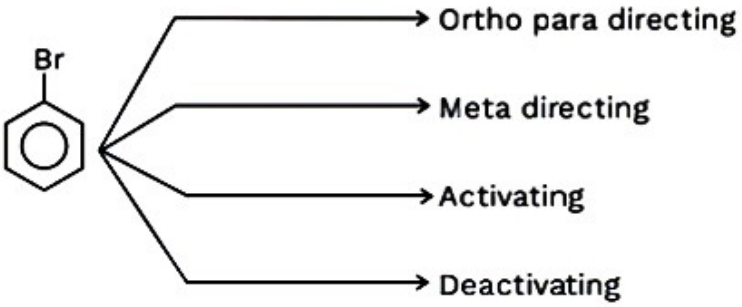
Q258



Sol. Ortho para directing (–I group),

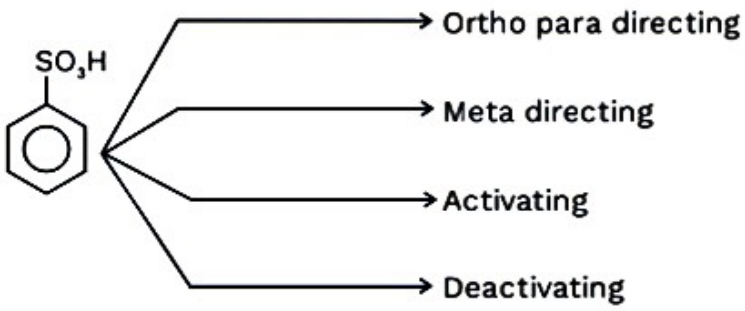
Deactivating (–x : are ortho/para directing)

Q259



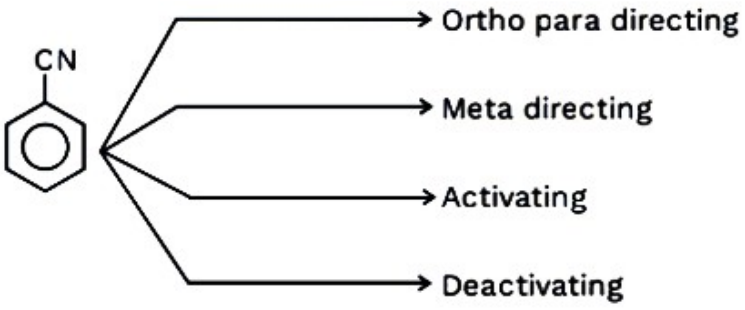
Sol. Ortho para directing , Deactivating

Q260



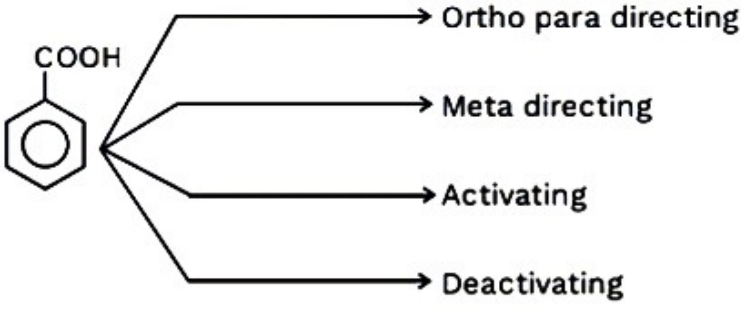
Sol. Meta director directing (Due to –M of SO_3H), Deactivating

Q261



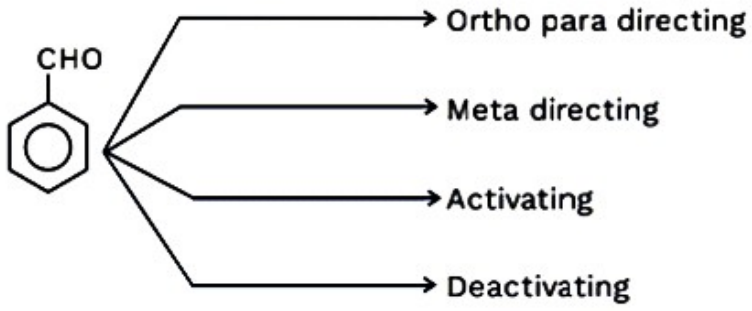
Sol. Meta director directing (–M of CN group), Deactivating

Q262



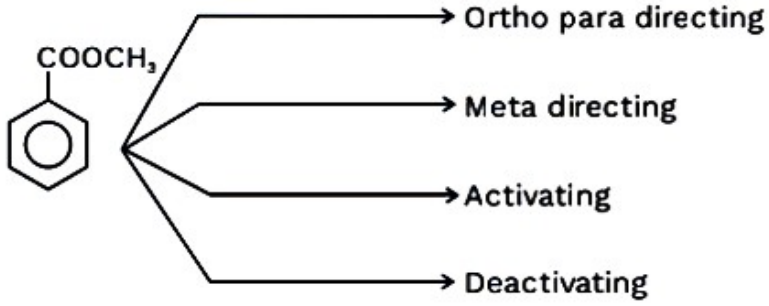
Sol. Meta director directing (–M of –COOH group), Deactivating

Q263



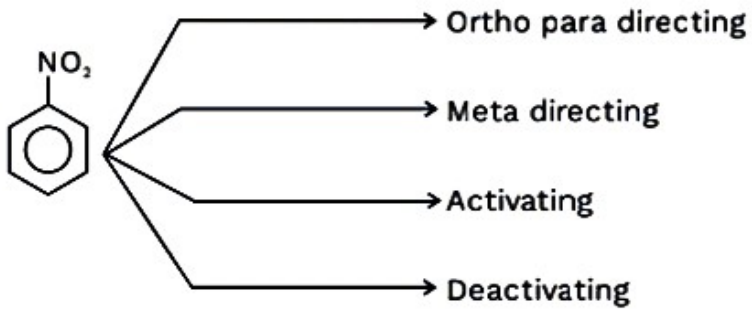
Sol. Meta director directing (–M of –CHO group), Deactivating

Q264



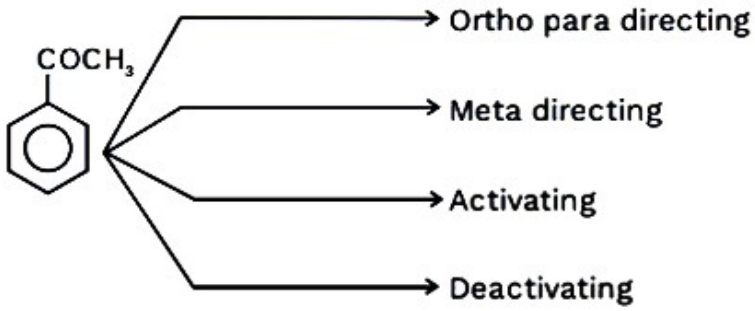
Sol. Meta directing (–M of $-COOCH_3$ group), Deactivating

Q265



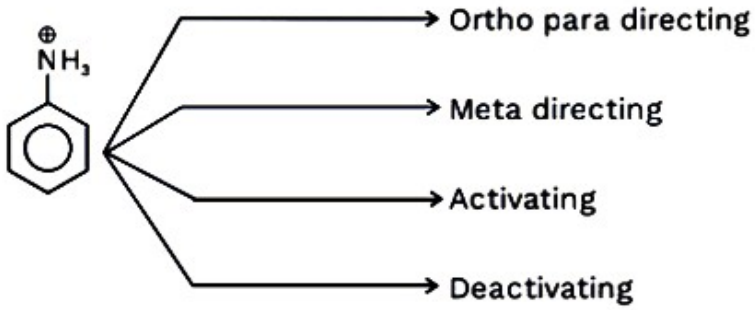
Sol. Meta directing (–M of $-NO_2$ group), Deactivating

Q266



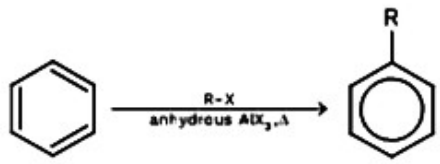
Sol. Meta directing (–M of $-\overset{\overset{O}{\|}}{C}-CH_3$ group), Deactivating

Q267

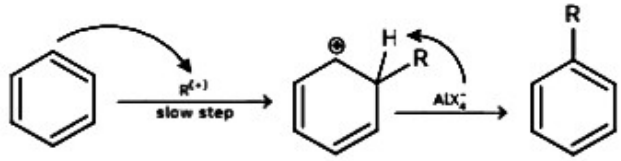


Sol. Meta directing (–M of $-\overset{\overset{O}{\|}}{C}-CH_3$ group), Deactivating

Friedel craft Alkylation :

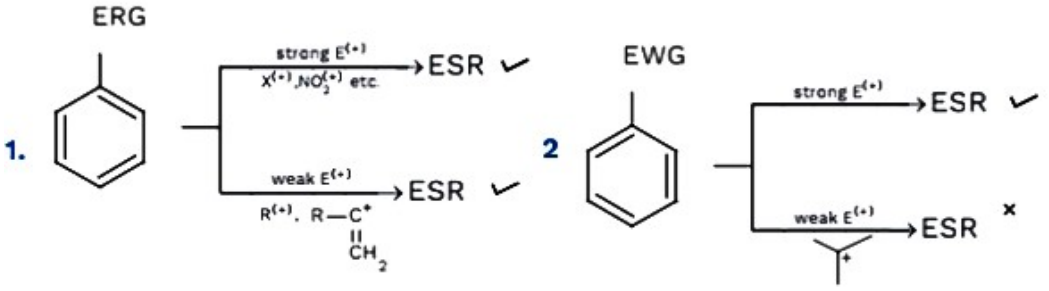


$$R - X + AlX_3 \rightarrow R^+AlX_4^-$$



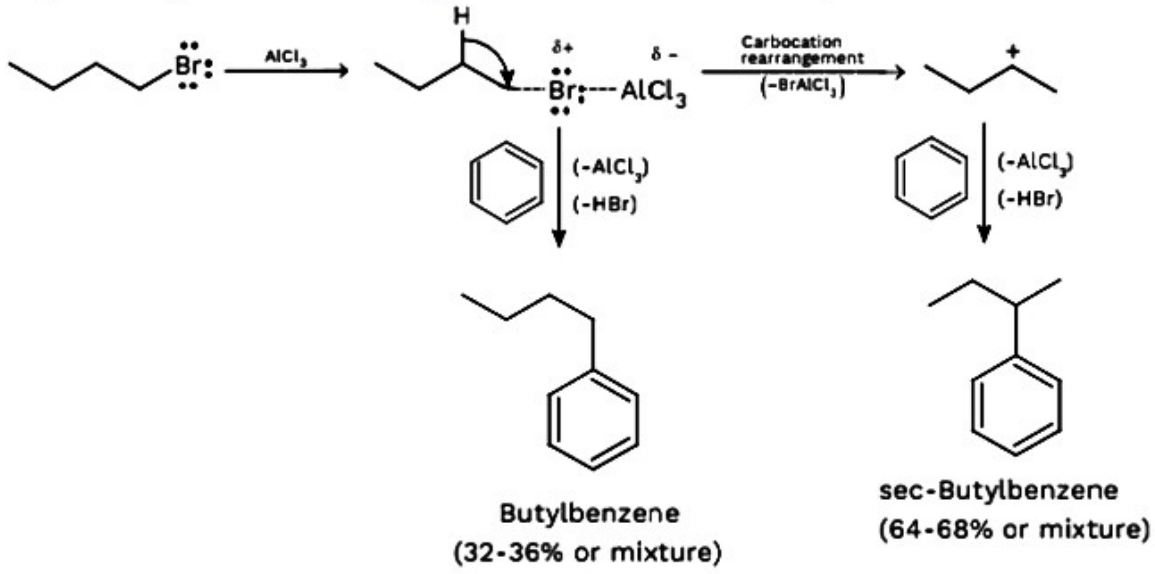
1. In this reaction weak electrophile form which is carbocation.
2. Rearrangement is possible so major product is decided by stable carbocation.
3. If strong e⁻ withdrawing groups is present then Friedel Craft alkylation and acylation is not possible because $R^{\oplus}$ and $R-\overset{\overset{O}{\|}}{C}{}^{+}$ are weak electrophile $E^{\oplus}$ and they could not interact with benzene due to strong withdrawing group.
4. In case of FeX_3, rearrangement is not possible. (as in FeX_3 pure carbocation does not form due to which rearrangement will not occur).

Concepts :

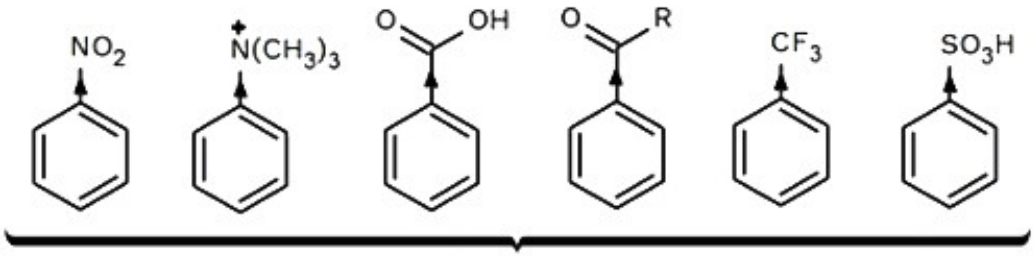


Limitations of Friedel-Craft Reactions

1. In Friedel craft Alkylation of Alkylhalide, Alkene or alcohol, Carbocation will form and it may rearrange to attain stability product will form with respect to stable carbocation.

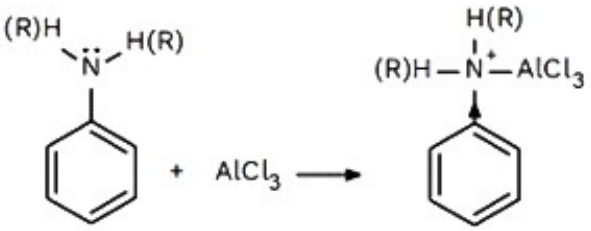


2. Friedel-Crafts alkylation and acylation reactions usually give poor yields when powerful electron-withdrawing groups are present on the aromatic ring.



These usually give poor yields in Friedel-Crafts reaction because the ring is electron deficient.

3. Aniline or benzene ring containing $-NR_2$, $-NHR$ group do not give Friedel-craft reaction because they from complex with Lewis Acid.

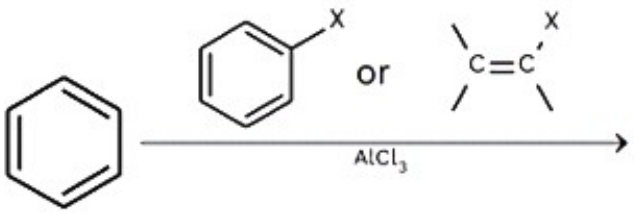


Does not undergo a Freidel-Crafts reaction

Points to remember

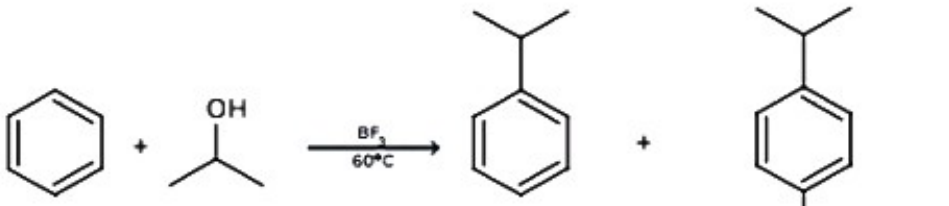
The groups like $-NH_2$, $-NR_2$, $-NHR$ in benzene ring by reacting with Lewis Acid convert into powerful electron withdrawing groups

4. Aryl Halide and vinylic halide do not give Friedel craft reaction because they form carbocation on sp^2 carbon which is not stable.



No Friedel-Crafts reaction because the halide is at an sp^2 carbon

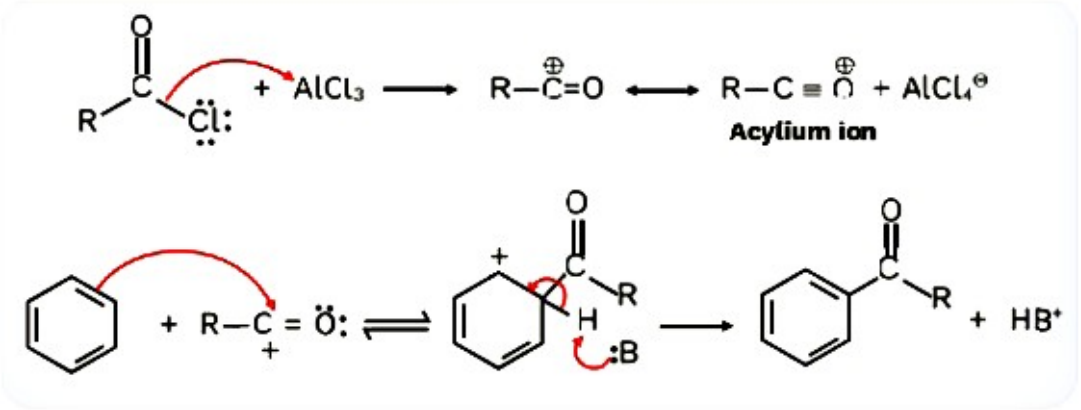
5. In case of polyalkylation, Alkyl group increase inductive effect so that as first alkylatin completed on benzene ring it be came more easy for further substitution:



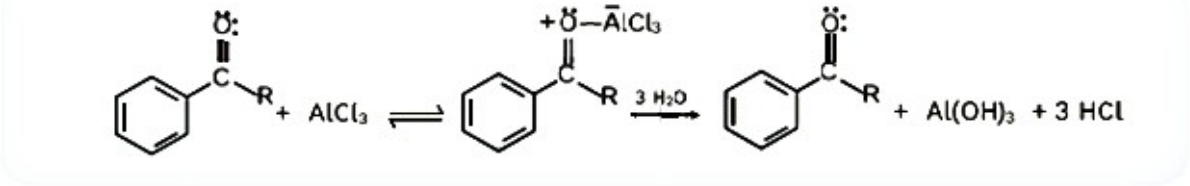
Isopropyl-benzene (24%)

p-Disoproxylbenzene (14%)

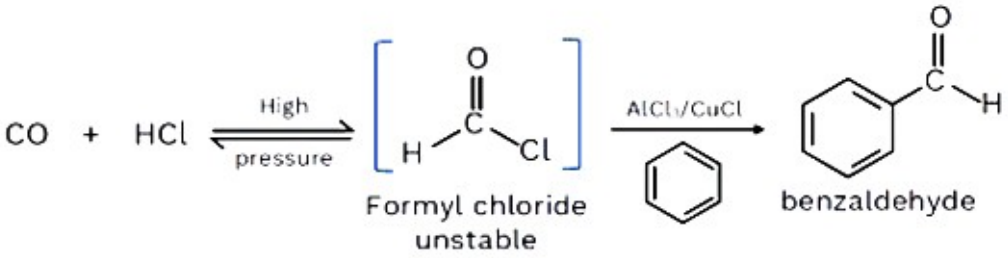
Mechanism for Friedel-Crafts acylation



Mechanism of fridel carft acylation with $AlCl_3$. After Acylation in third step water should be added to reaction mixture for product formation from complex.

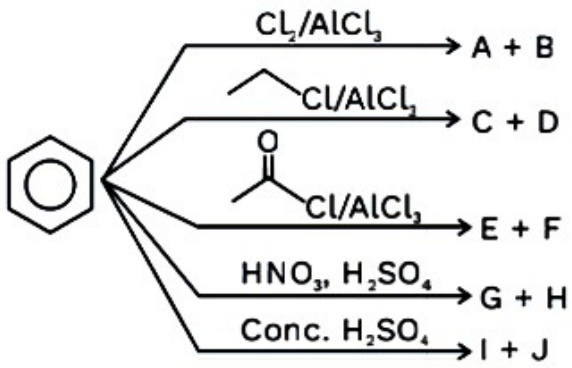


To formation of benzaldehyde from benzene is not proceed with formyl chloride because Acyl Halide used for this reaction must be unstable so that it can form stable carbocation, while in case of formeyl chloride, Gatterman-Koch formylation reaction is used which is at high pressure and occurs in presence of carbon monoxide and HCl along with aluminum chloride – Cuprous Chloride Catalyst give formylation.

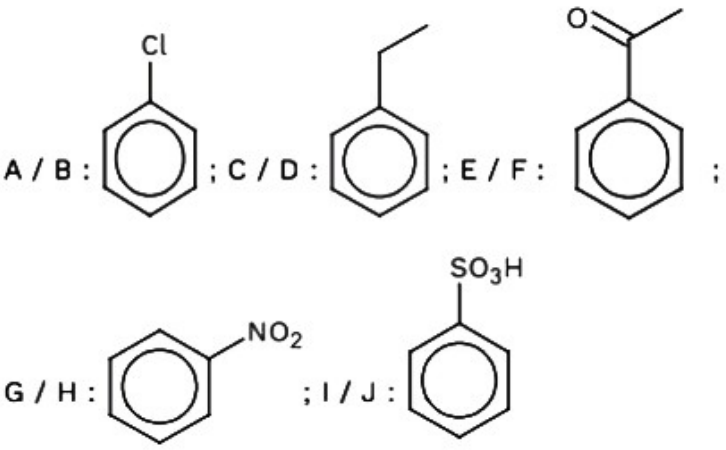


Q. From the given compounds find out rate of electrophilic aromatic substitution

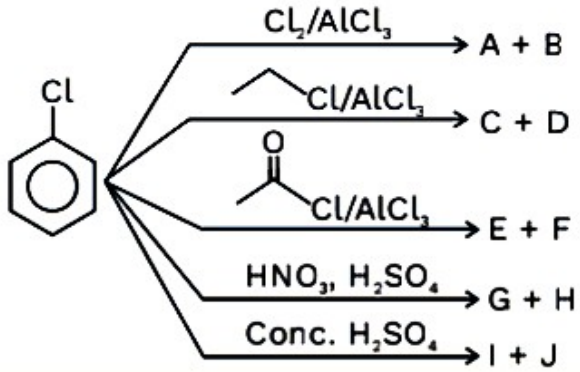
Q268



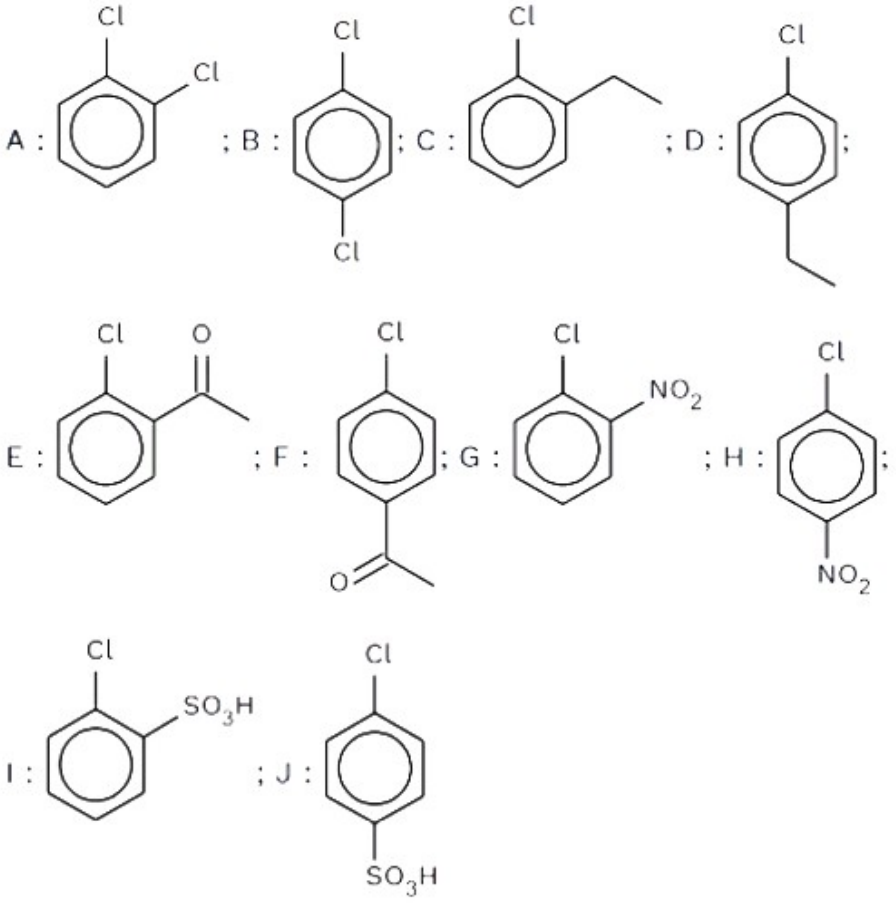
Sol.



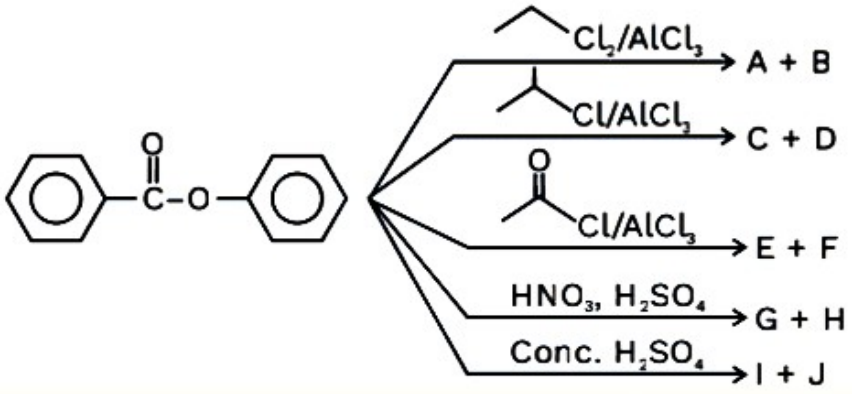
Q269


Cl
$Cl_2/AlCl_3$
A + B
Cl/$AlCl_3$
C + D
O
Cl/$AlCl_3$
E + F
HNO_3, H_2SO_4
G + H
Conc. H_2SO_4
I + J

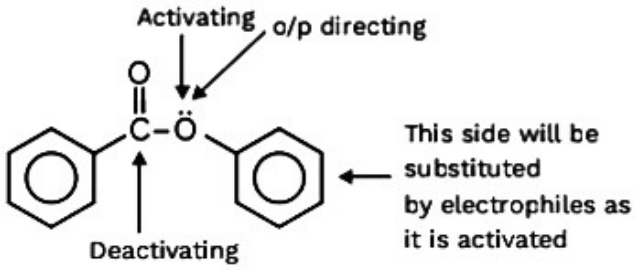
Sol.

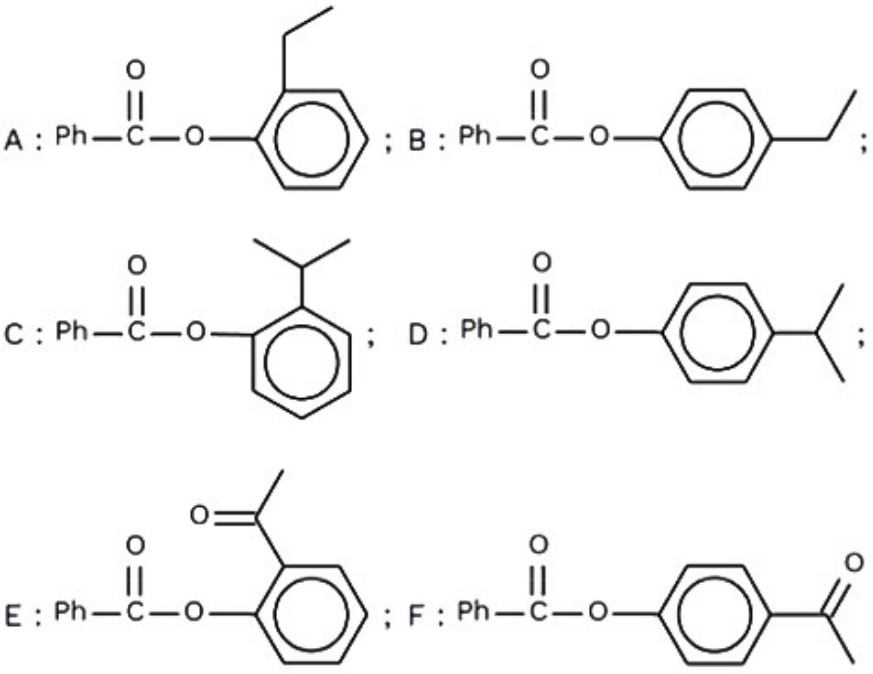

A :
Cl
Cl
; B :
Cl
Cl
; C :
Cl
; D :
Cl
;
E :
Cl
O
; F :
Cl
O
; G :
Cl
NO_2
; H :
Cl
NO_2
;
I :
Cl
SO_3H
; J :
Cl
SO_3H

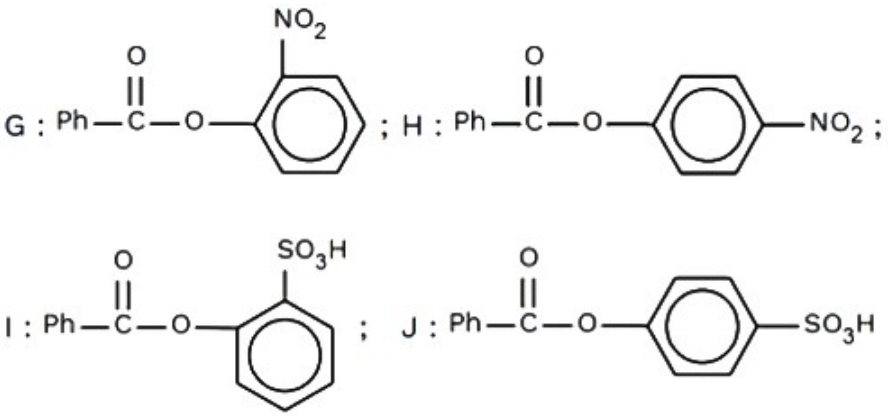
Q270


$Cl_2/AlCl_3$
A + B
Cl/AlCl$_3$
C + D
Cl/AlCl$_3$
E + F
HNO_3, H_2SO_4
G + H
Conc. H_2SO_4
I + J

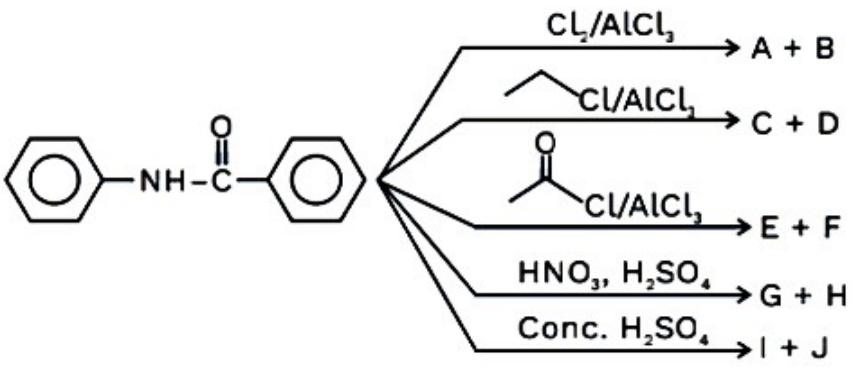
Sol.


Activating
o/p directing
This side will be substituted by electrophiles as it is activated
Deactivating

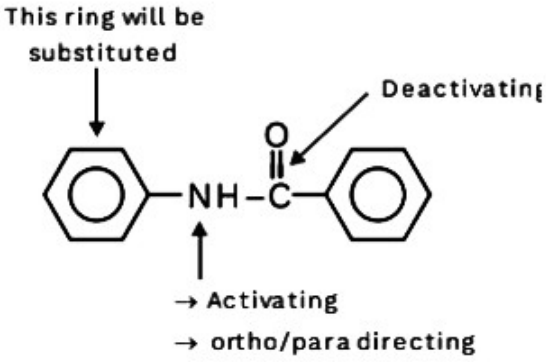

A : Ph—C—O
; B : Ph—C—O
;
C : Ph—C—O
;
D : Ph—C—O
;
E : Ph—C—O
; F : Ph—C—O
;

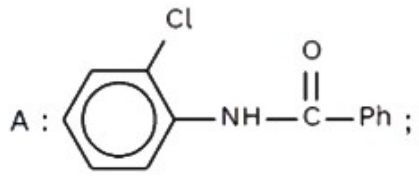

G : Ph—C—O
NO2
; H : Ph—C—O
NO2 ;
I : Ph—C—O
SO3H
; J : Ph—C—O
SO3H

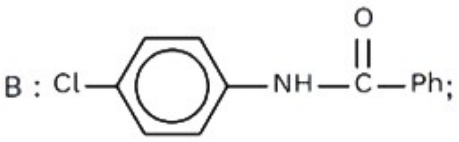
Q271

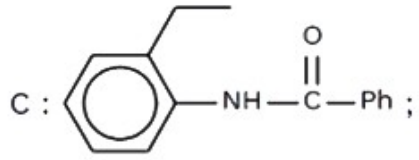

NH–C
Cl2/AlCl3
A + B
Cl/AlCl3
C + D
Cl/AlCl3
E + F
HNO3, H2SO4
G + H
Conc. H2SO4
I + J

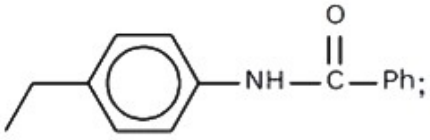
Sol.

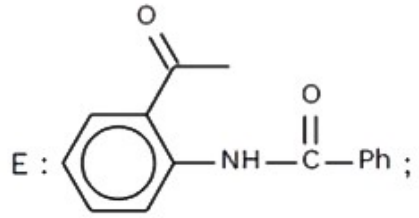

This ring will be substituted
Deactivating
NH–C
→ Activating
→ ortho/para directing

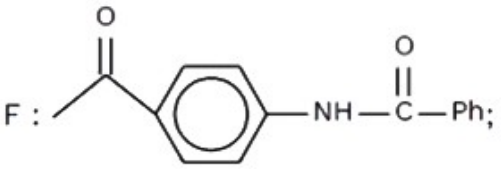
A : 

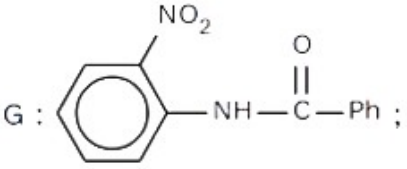
B : 

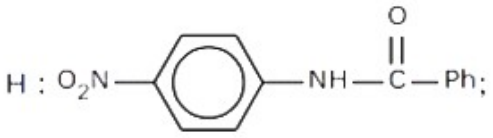
C : 

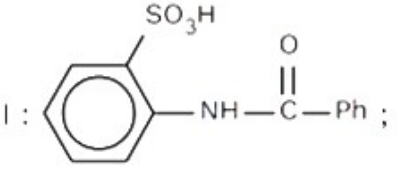
D : 

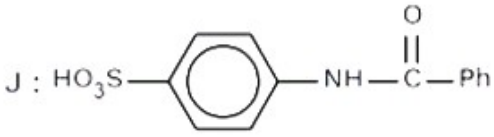
E : 

F : 

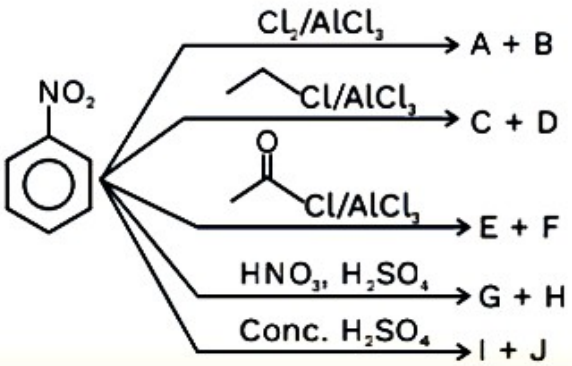
G : 

H : 

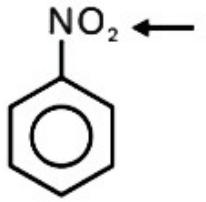
I : 

J : 

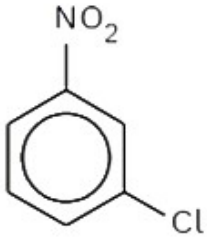
Q272

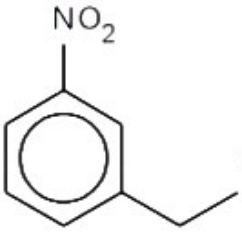


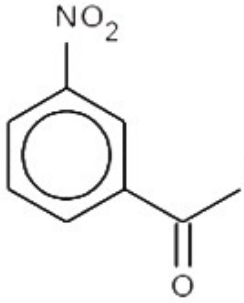
Sol.



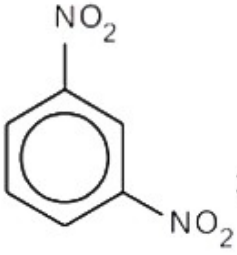
← Meta directing group
Also deactivating
⇒ Low yield of product formation

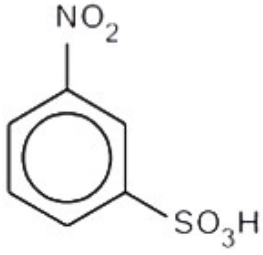
A / B : 

; C / D : 

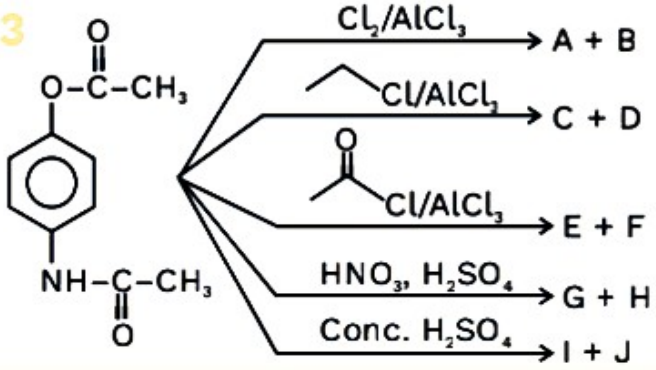
; E / F : 

;

G / H : 

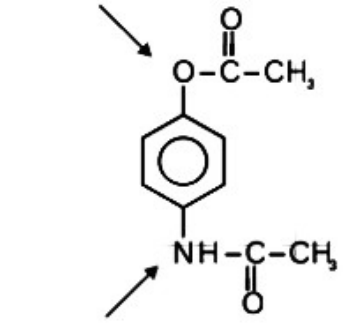
; I / J : 

Q273



Sol.

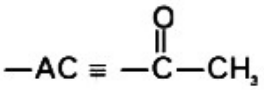
Activating
Ortho/para directing

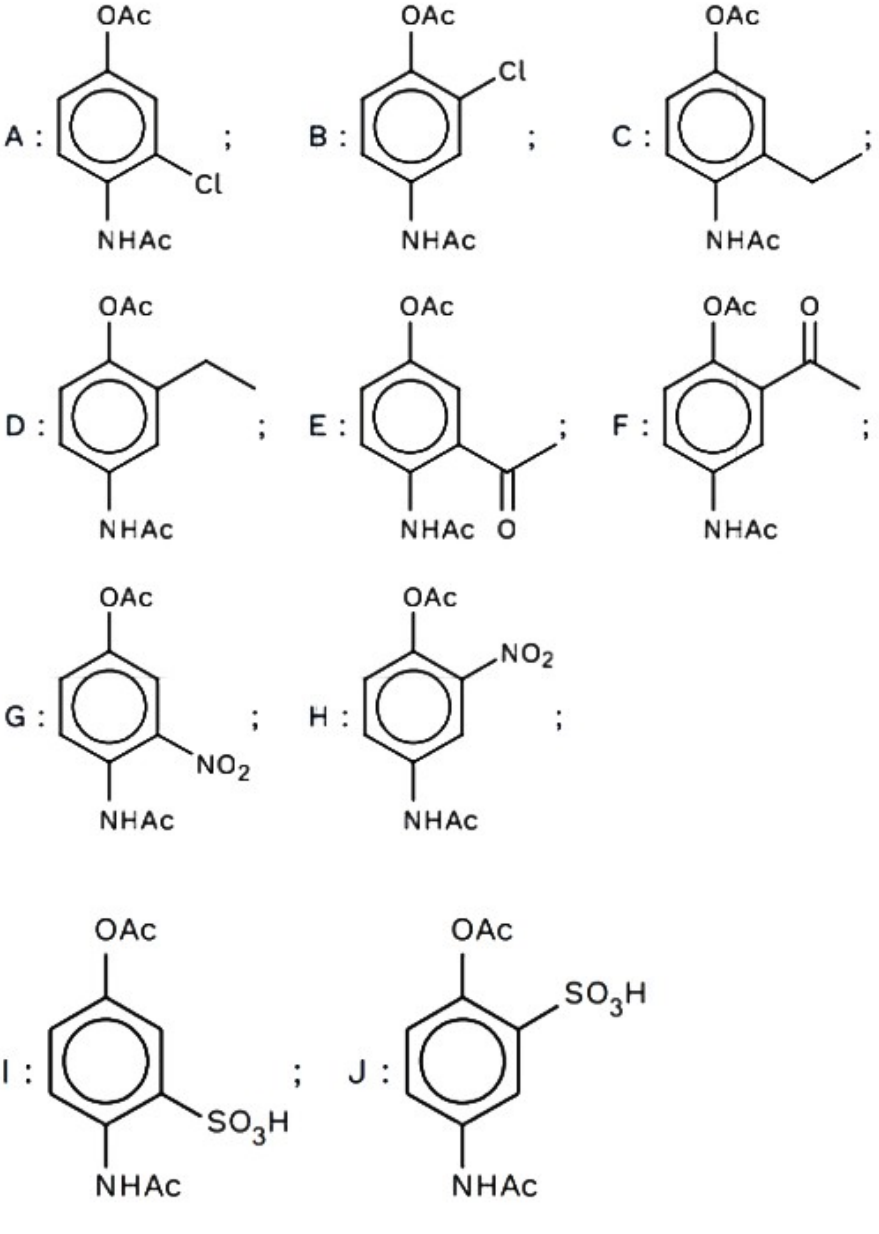


Activating
Ortho/para directing

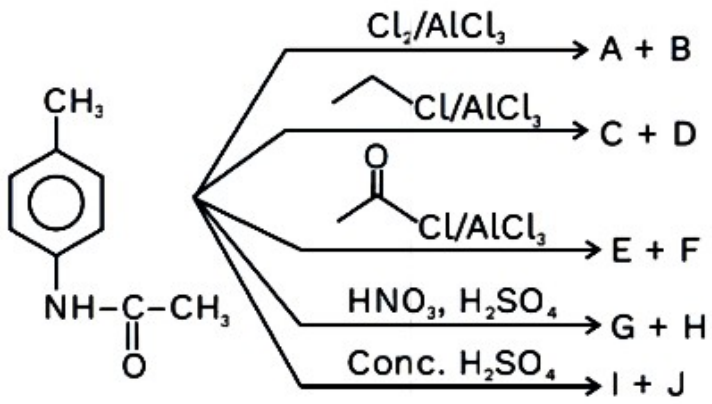
→ As –NH– is more activating then –O– due to strong +m effect, Substitution will take place w.r.t –NH– for major product

→ As para position is already blocked only ortho product will be formed

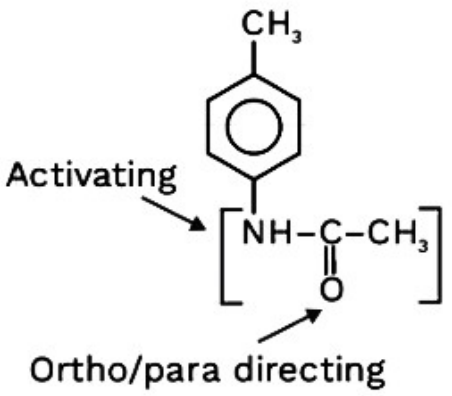




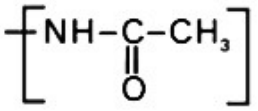
Q274



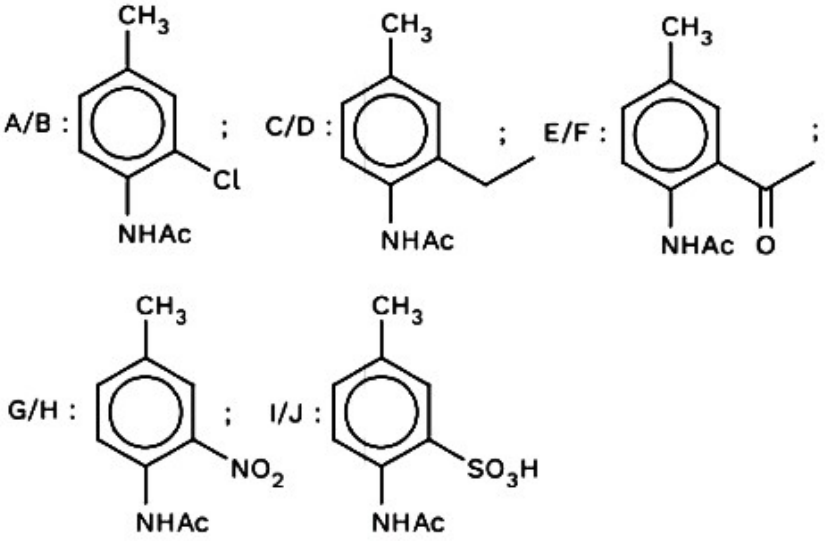
Sol.



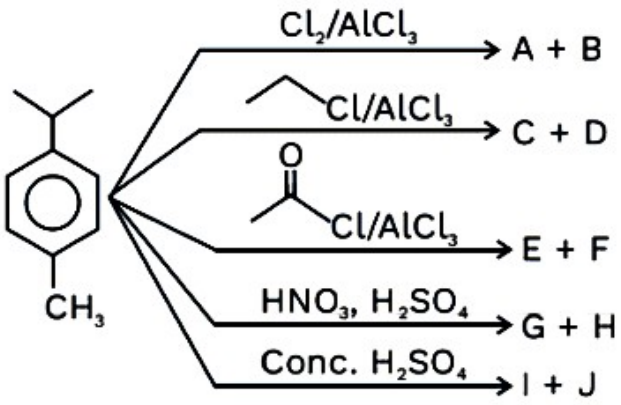
→ Products will be formed by substitution of electrophiles at ortho position w.r.t



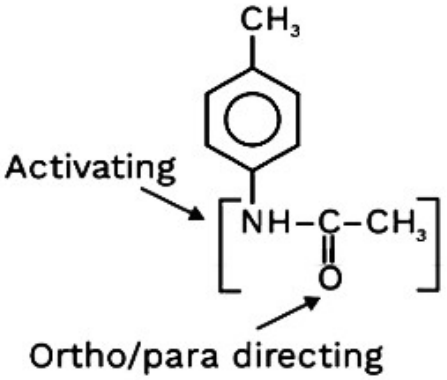
→ P ara position is blocked so no substitution is possible at ortho



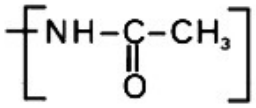
Q275



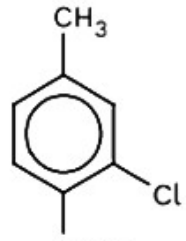
Sol.

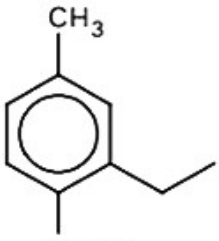


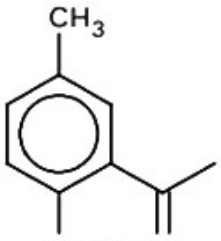
→ Products will be formed by substitution of electrophiles at ortho position w.r.t



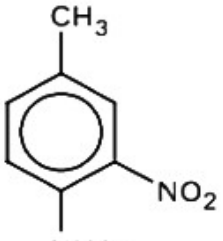
→ P ara position is blocked so no substitution is possible at para

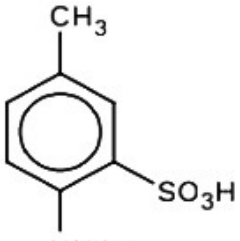
A / B : 

; C / D : 

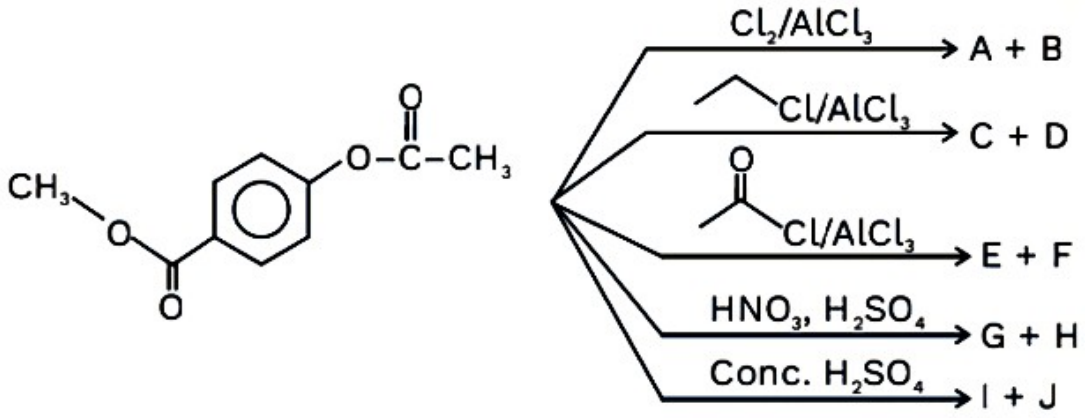
; E / F : 

;

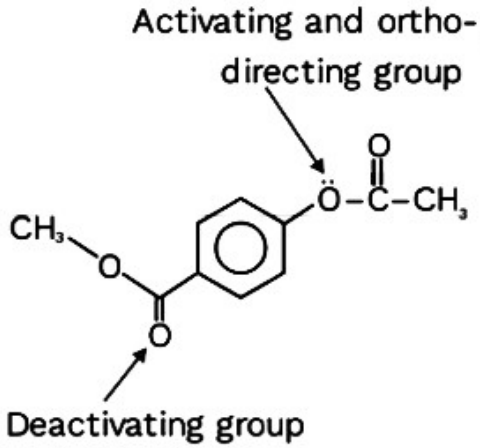
G / H : 

; I / J : 

Q276



Sol.



→ Products will be formed by substitution of ($E^{\oplus}$) at ortho w.r.t $-O-\overset{O}{\overset{\|}{C}}-CH_3$

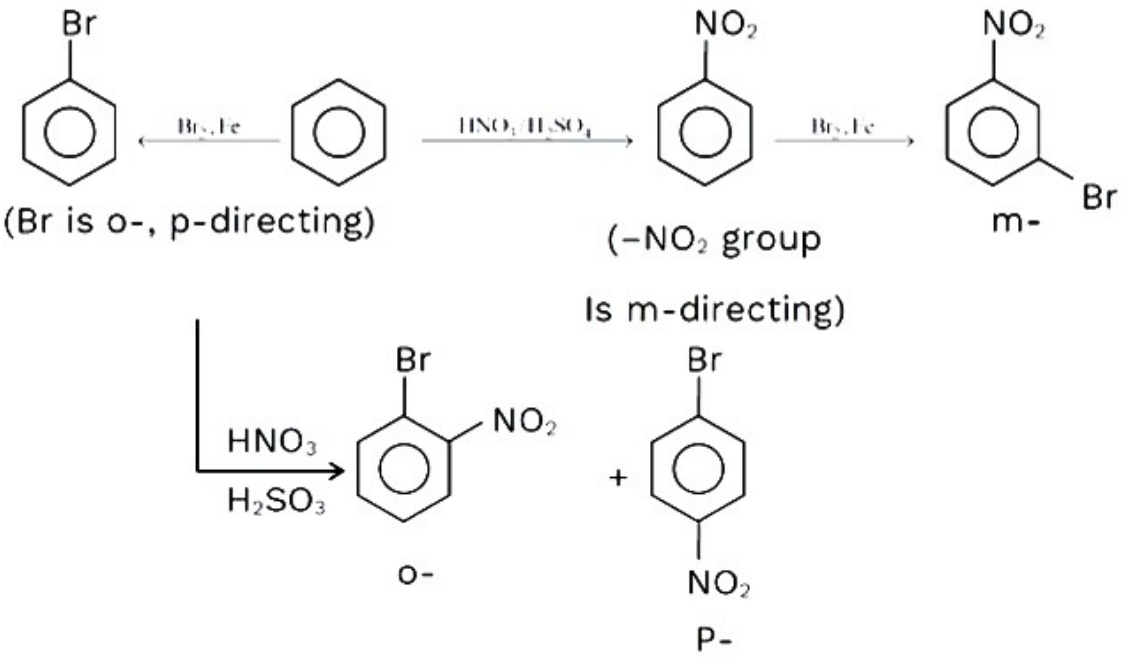
→ Substitute $Cl^{\oplus}$, $CH_3CH_2^{\oplus}$, $CH_3-\overset{O}{\overset{\|}{C}}-^{\oplus}$, $\overset{\oplus}{N}O_2$ and $\overset{\oplus}{S}O_3H$

[At ortho position w.r.t $-O-\overset{O}{\overset{\|}{C}}-CH_3$] formation of A/B, C/D/, E/F, G/H and I/J respectively.

Orientation and Synthesis

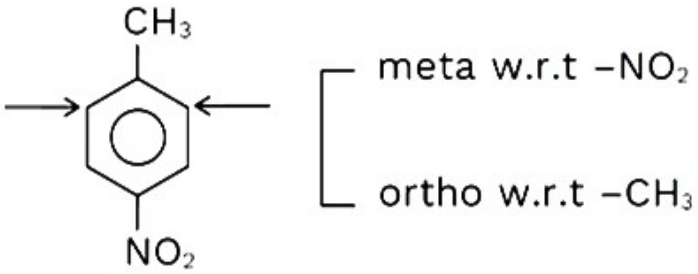
By using the nature of directing group (substituent), it is possible to prepare variety of products.

To prepare meta-bromo nitrobenzene from benzene, we have to first carry out nitration and then bromination ; and to prepare o-and p-derivatives, sequence will be bromination and then nitration.

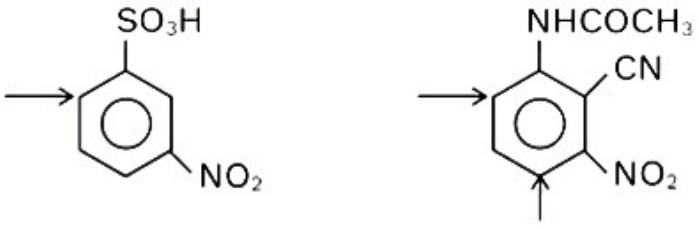


Orientation in Disubstituted benzene

When two substitution are present, orientation of the third coming SE is complicated. $-CH_3$ group is o–, p-directing and $-NO_2$ group is m-directing, thus in p nitrotoluene, new electrophile can be in a position shown by arrow:



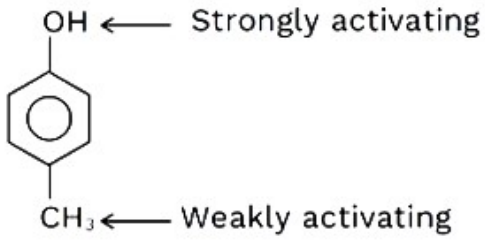
Similarly we can decide the position of new electrophile in other disubstituted compounds.



both meta-directing

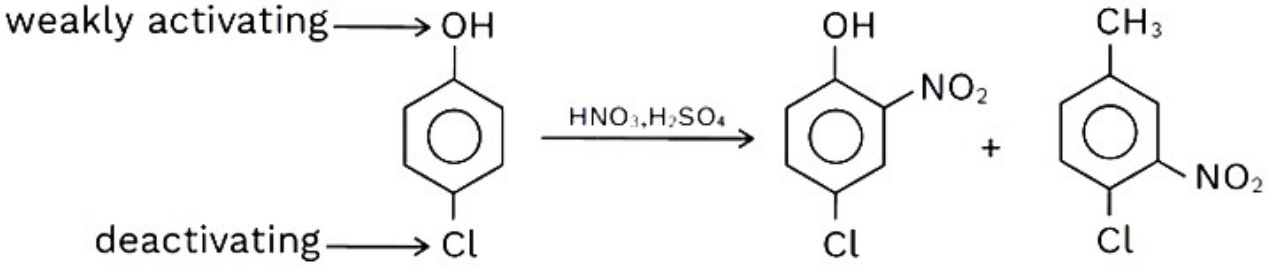
$NHCOCH_3$ ortho, para and –CN meta-directing

Problems may arise in the following cases :

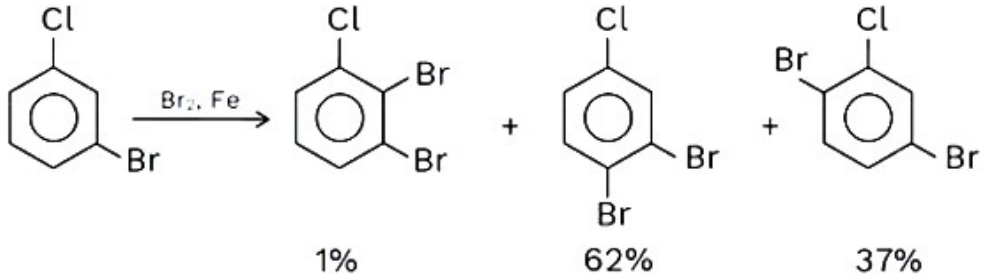


Both are ortho, para-directing and none of the vacant positions are ortho - and para, w.r.t. each. In such cases we follows :

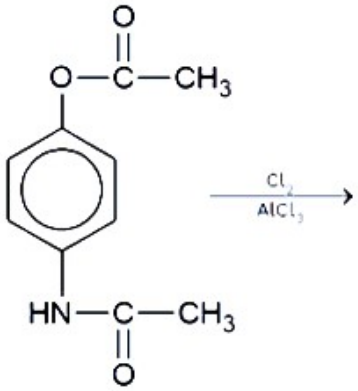
(a) Strongly activating groups generally win over deactivating or weakly activating groups.



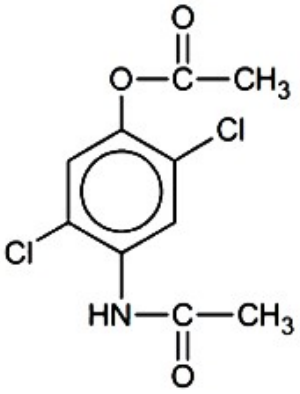
(b) If the substitution is meta to each other, S_E in between vacant sites takes place rarely.



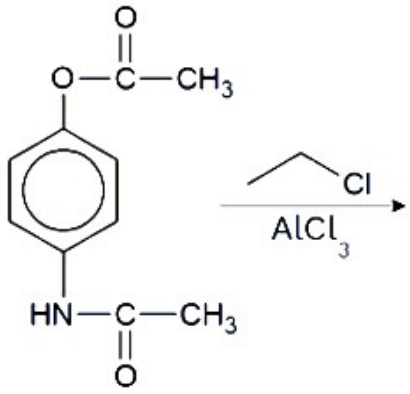
Q277



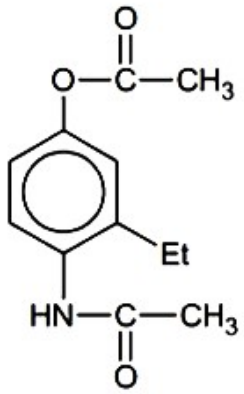
Sol.



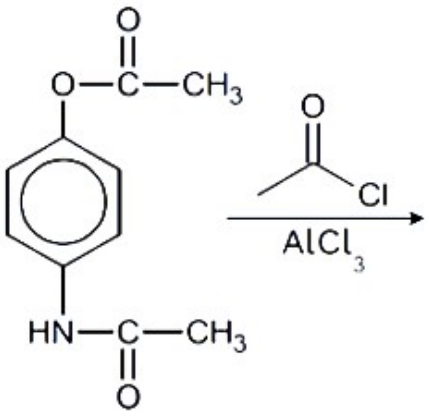
Q278



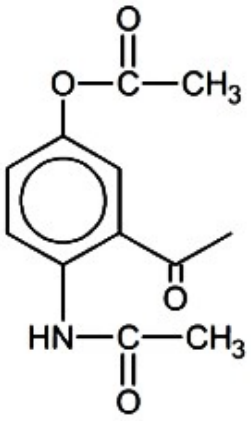
Sol.



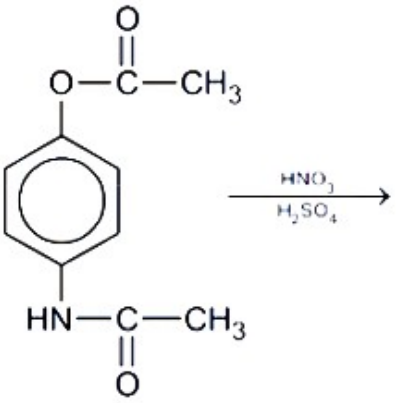
Q279



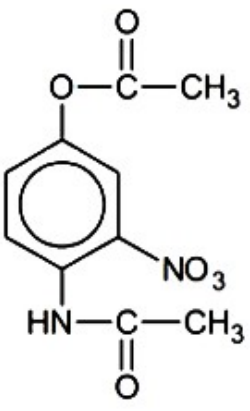
Sol.



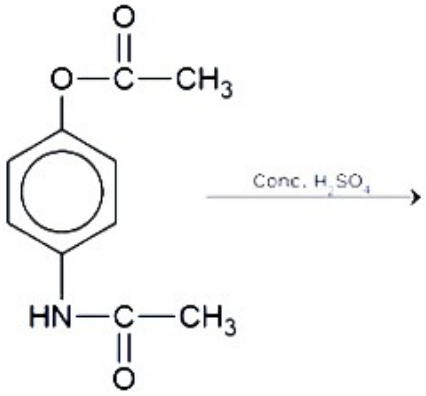
Q280



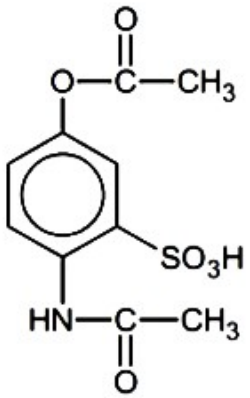
Sol.



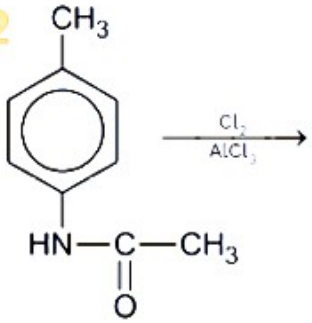
Q281



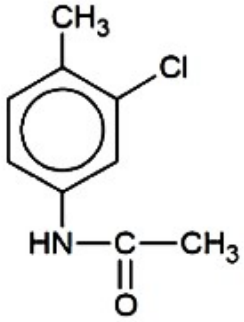
Sol.



Q282

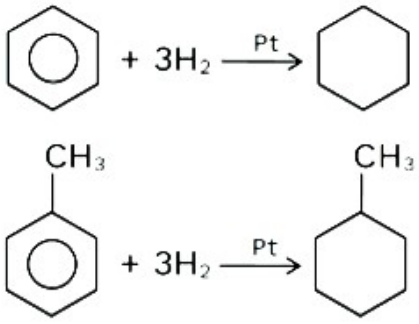


Sol.

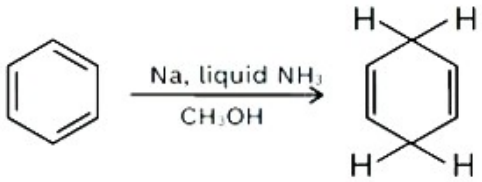


N.C.E.R.T. Reactions

Catalytic hydrogenation

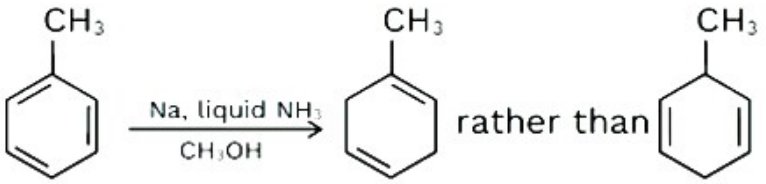


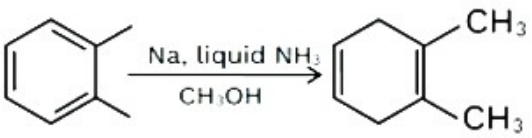
Treatment of arenes with group I metal (like sodium, lithium) and methanol (or ethanol) in liquid reduces arenes to non-conjugated dienes.



1,4-cyclohexadienes

Alkyl substituted arenes give 1, 4-cyclohexadienes in which the alkyl group is a substituent on the double bond.

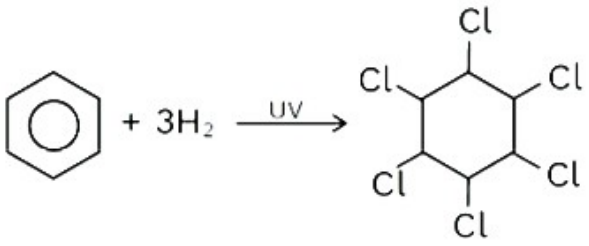




Metal ammonia -alcohol reduction of aromatic ring is known as Birch reduction and is one type of a more general class of reactions called dissolving metal reduction.

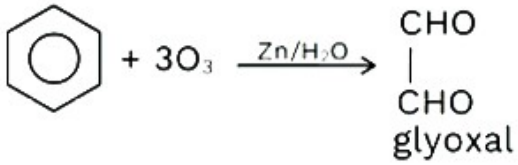
Addition of Halogen

Benzene hexachloride (BHC, $C_6H_6Cl_6$) is also called Lindane or Gammaxene (γ-isomer) and is used insecticide.

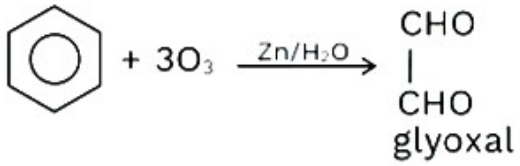


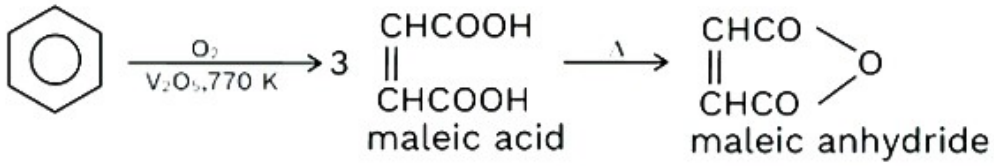
benzenehexachloride, B.H.C, 666

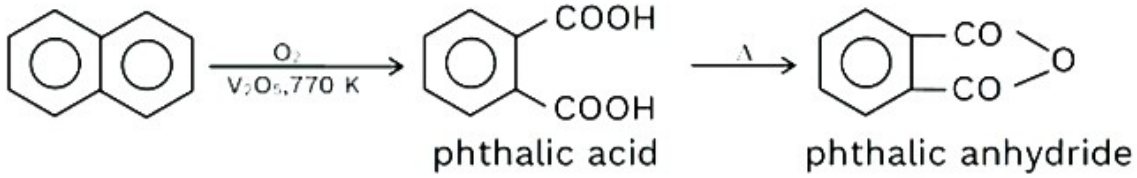
Ozonolysis

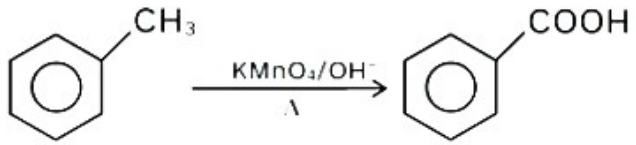


Oxidation

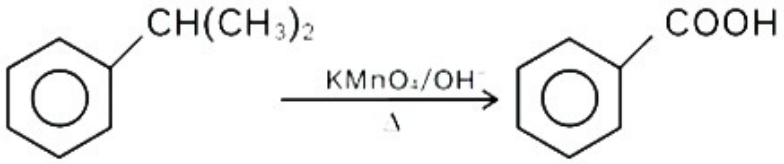




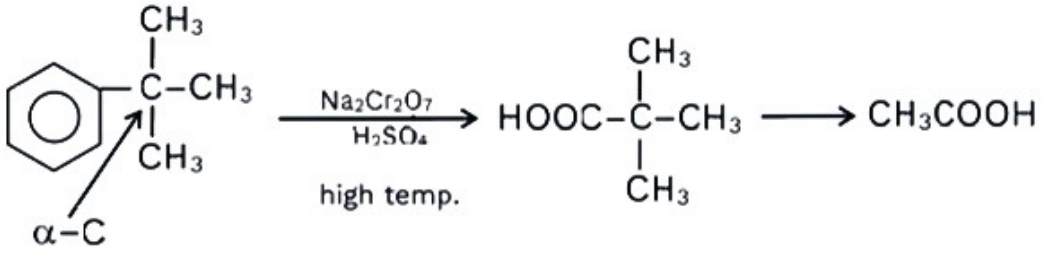


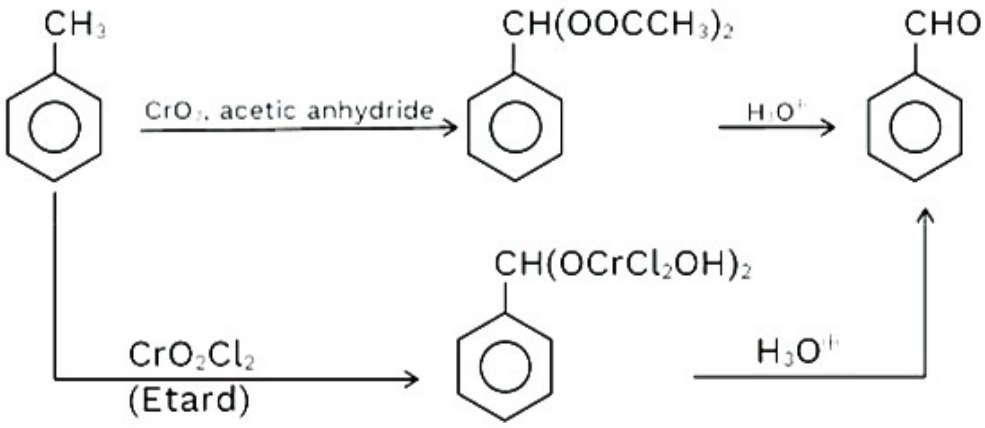


When oxidised using$KMnO_4/OH^-$,$Na_2Cr_2O_7/H+$,$KMnO_4/H^+$ the entire side chain, with at least one H at α-C, regardless of length is oxidised to –COOH.

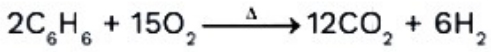


The conversion of alkyl benzenes to benzoic acid with strong oxidations suggest that benzene ring is more stable than the side chains ; this is correct so long as the side-chain contains at least one benzylic C–H bond. The first step in the mechanism is the removal of hydrogen from the benzylic carbon. If the side-chain has no C–H bond at the α-carbon (w.r.t. benzene nucleus), then benzene ring is cleaved.

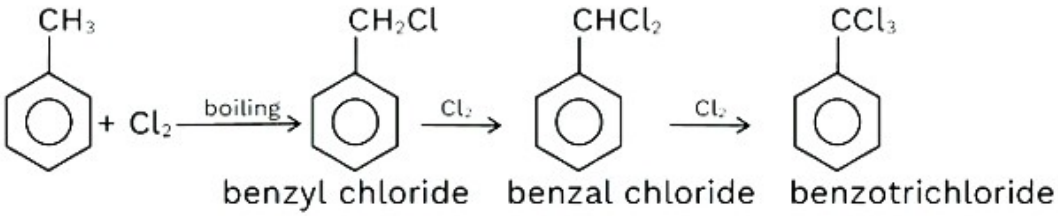




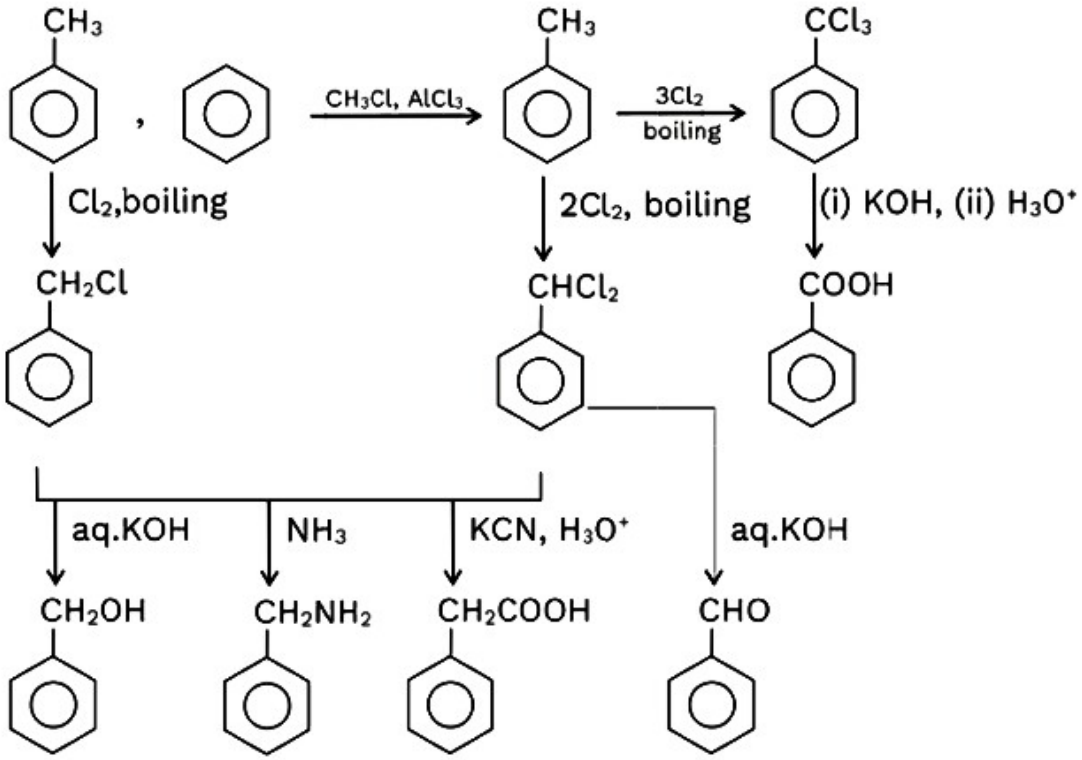
Combustion



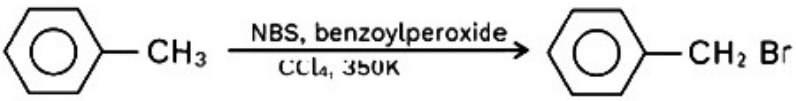
Side-chain halogenation



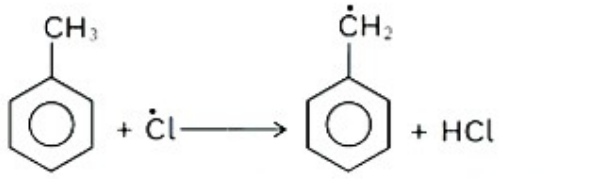
This provides a source of preparing other derivatives from benzene and toluene. Example:



Example:



(NBS is N-bromo succinimide)

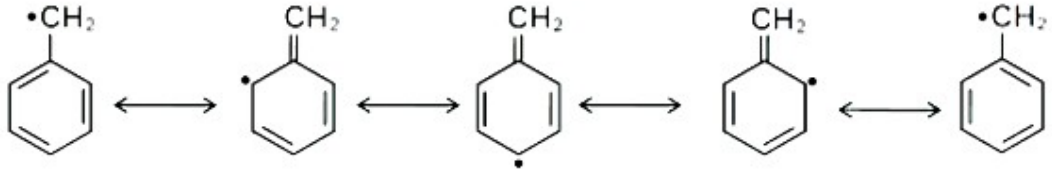


Stable benzyl free radical

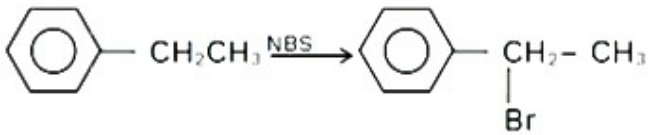
Points to remember

Side-chain halogenation involves free radical mechanisms due to lower bond energy of the benzyl C–H bond :

Benzyl radical is stabilised by resonance due to delocalisaton of odd electrons into ring :

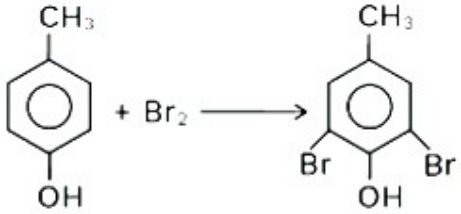


NBS can also be used for bromination of side-chain alkyl group :



(by allylic substitution)

In presence of an activating group–OH or $-NH_2$, side chain halogenation is difficult:



Combustion

Combustion is highly exothermic, due to this high amount of heat evolved, these hydrocarbons (Alkanes) used as fuel while incomplete combustion of Alkane give carbon black which is used to produce ink, blank pigments and as a filters.

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(1); \qquad \Delta_c H^\circ = -890 \text{ kJ mol}^{-1}$$

$$C_4H_{10}(g) + \frac{13}{2}O_2(g) \rightarrow 4CO_2(g) + 5H_2O(1); \qquad \Delta_c H^\circ = -2875.84 \text{ kJ mol}^{-1}$$

Combustion equation:

$$C_nH_{2n+2} + \left(\frac{3n+1}{2}\right)O_2 \rightarrow nCO_2 + (n+1)H_2O$$

In presence of air (oxygen), Alkene oxidized into carbondioxide and water on Heating and reaction is highly exothermic.

Controlled oxidation

Alkanes give various oxidation products when heating with controlled supply of oxygen or air at high pressure and in presence of catalyst.

(i) $2CH_4 + O_2 \xrightarrow{Cu/523K/100atm} \underset{\text{Methanol}}{2CH_3OH}$

(ii) $CH_4 + O_2 \xrightarrow[\Delta]{Mo_2O_3} \underset{\text{Methanal}}{HCHO} + H_2O$

(iii) $2CH_3CH_3 + 3O_2 \xrightarrow[\Delta]{(CH_3COO)_2Mn} \underset{\text{Ethanoic acid}}{2CH_3COOH} + 2H_2O$

(iv) Tertiary Hydrogen containing Alkanes can be oxydise in respective alcohols by potassium permanganate while other alkanes resist oxidation.

$$\underset{\text{2-Methylopropane}}{(CH_3)_3CH} \xrightarrow[\text{Oxidation}]{KMnO_4} \underset{\text{2-Methylpropan-2-ol}}{(CH_3)_3COH}$$

Isomerisation

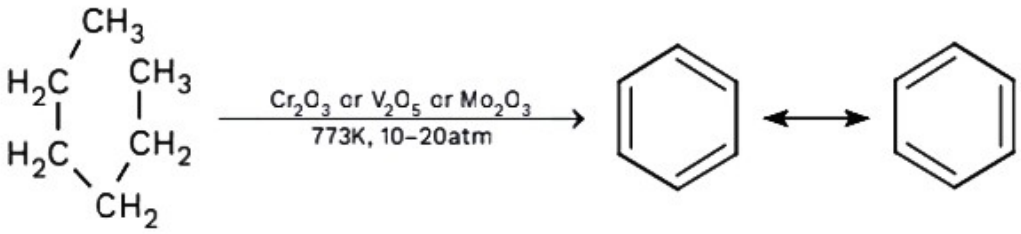
In presence of Anhydrous Aluminum Chloride and HCl, normal Alkane isomerise on heating and give branched alkane.

$$\underset{\text{n-Hexane}}{CH_3(CH_2)_4CH_3} \xrightarrow{\text{Anhy. } AlCl_3/HCl} \underset{\text{2-Methylpentane}}{CH_3CH(CH_3)-(CH_2)_2-CH_3} + \underset{\text{3-Methylpentane}}{CH_3CH_2-CH(CH_3)-CH_2-CH_3}$$

Aromatization

In presence of oxides of vanadium, moly b denum or chromium at high temperature 773K and 10-20 atm pressure give aromatic ring of corresponding n-Alkane, this is due to dehydrogenation of Alkane.

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Toluene (C_7H_8) is methyl derivative of benzene. Which alkane do you suggest for preparation of toluene?

Reaction with steam

In presence of Ni-catalyst at 1273K temperature methane react with steam give carbon monoxide and Hydrogen gas. This also industrial method to preparation of hydrogen.

Pyrolysis

At high temperature, higher alkanes decompose into lower alkanes on heating. The decomposition by application of heat is called pyrolysis or cracking.

Polymerisation

(a) Linear Polymerisation

In this reaction two or more alkene unites together in such a way to give a higher molecular weight which is multiple of the original alkene. This process is called Polymerisation. Polymerisation occurs under suitable condition due to linear Alkyne combined and give high molecular weight polyene and called as linear polymerisation these polymers are used as electrodes in batteries due to conducts electricity and cheaper, lighter compare then metals.

Example

$$n\ \ CH_2 = CH - CH = CH_2 \xrightarrow[\text{\& temperature}]{\text{High pressure}} (-CH - CH - CH - CH -)_n$$

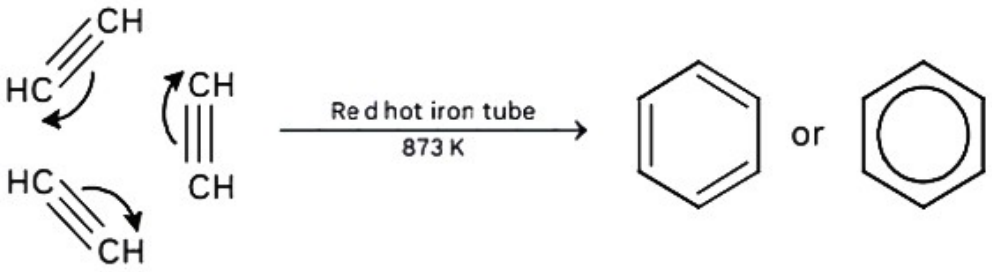
Polymer

(b) Cyclic Polymerisation

When Ethyne is passing through red hot tube give cyclic polymerization at 873K temperature. By this method Aliphatic Compounds Converts into aromatic compound for examples there ethyne molecules give benzene by polymerization.

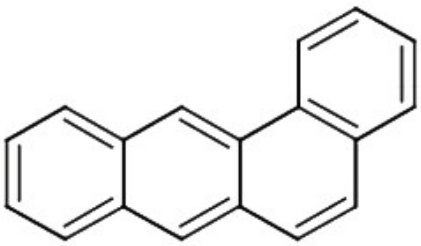
⇒ It is used to prepare various derivatives of benzene.

⇒ It is used to prepare dye and drugs and other organic compounds.

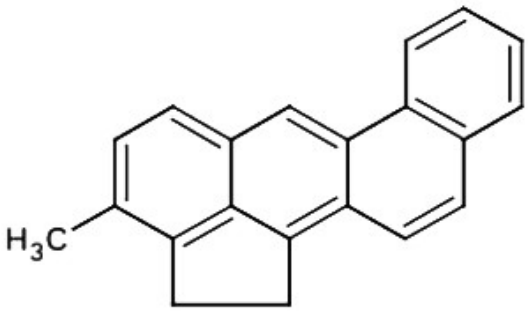


Carcinogenicity and Toxicity

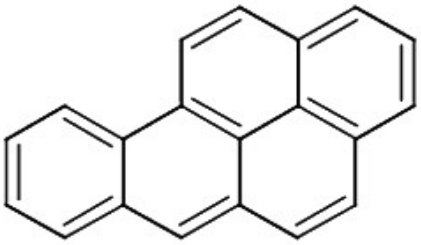
Carcinogenicity and toxicity organic molecules are those in which two or more benzene rings are fused together or they are also form by polynuclear hydrocarbons. These are toxic due to cause of cancer producing property because they do biochemical reaction in human body and damage DNA.
Examples - Tobacco, petroleum, coal etc.



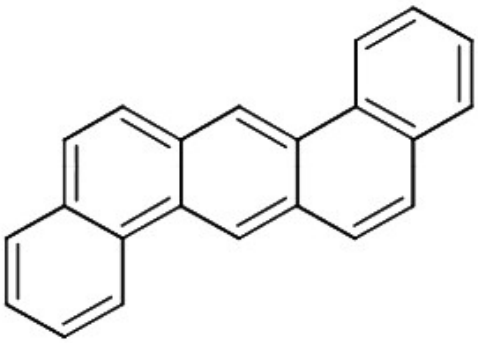
1,2-Benzanthracene



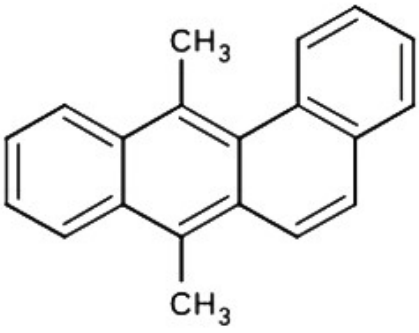
3-Methylcholanthrene



1,2-Benzanthracene



1,2,5,6-Dibenzanthracene



9,10-Dimethyl-1,2-Benzanthracene